ธนาคารแห่งประเทศไทย
BANK OF THAILAND



Monetary Policy Report

March

2020

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2020

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC implements monetary policy under the flexible inflation targeting regime. While regarding medium–term price stability as its primary objective, the MPC also aims at supporting sustainable, full-potential economic growth and preserving financial stability, attributing to long-term price stability and economic sustainability

Monetary Policy Target

On December 24, 2019, the Cabinet approved the monetary policy target for 2020, which was mutually agreed between the MPC and the Minister of Finance to set the headline inflation within the range of 1–3 percent as the target for the medium-term horizon and for 2020.

In the event that average headline inflation in the past 12 months or a forecast of average headline inflation over 12 months ahead breaches the target range, the MPC shall send an open letter to the Minister of Finance to explain reasons for the breach of the target range, together with measures taken and estimated time to bring inflation back to the target. In addition, the MPC will write an additional open letter to the Minister of Finance every six months if average headline inflation based on the above criteria remains outside the target range.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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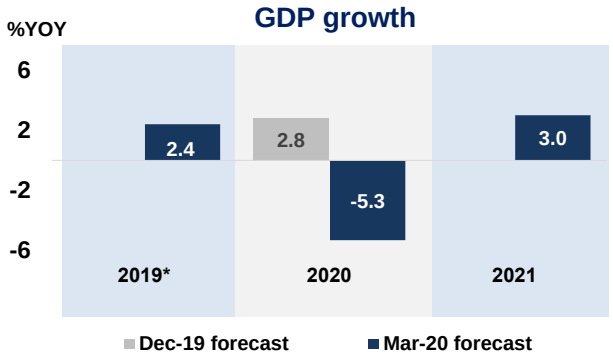
Economic assessment

Financial stability assessment

Note: Data in this report is as of March 24, 2020 (one day before the Monetary Policy Committee meeting)

Key Highlights of the Monetary Policy Report, March 2020

Economic growth projection

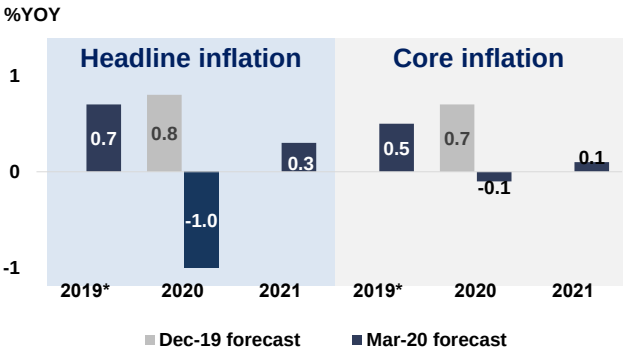


Note: • Outturn
Source: NESDC, estimated by Bank of Thailand

Key factors

- COVID-19 outbreak
- More severe impact of the drought
- Government's relief measures

Inflation projection

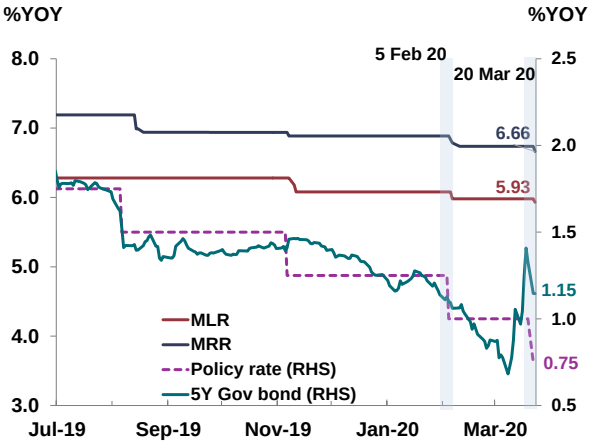


Note: • Outturn
Source: Ministry of Commerce, estimated by Bank of Thailand

Key factors

- Lower demand in line with economic conditions
- Lower crude oil prices

Financial conditions

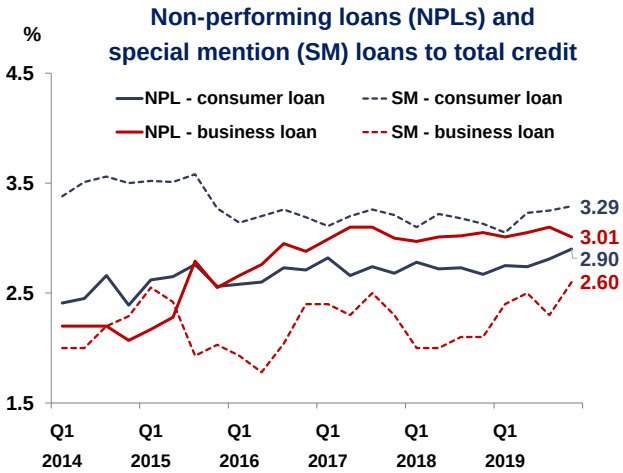


Source: ThaiBMA and Bank of Thailand (data as of 24 March 2020)
Note: MLR and MRR are averages of 5 largest commercial banks' lending rates

Key factors

- Policy rate was at historically low level
- MLR and MRR continued to fall
- Measures to stabilize the financial markets (22 Mar 20) helped curb volatilities in the bond markets

Financial stability



Source: Bank of Thailand (data as of end of Q4 2019)

Key factors

- NPLs and SM loans were expected to rise
- Debt servicing ability of businesses and households was expected to deteriorate due to income shocks

Executive Summary

Monetary Policy Conduct in the First Quarter of 2020

At the meeting on February 5, 2020, the Committee voted unanimously to cut the policy rate by 0.25 percentage point from 1.25 to 1.00 percent. The Committee assessed that the Thai economy would expand at a much lower rate than the previous forecast and much further below its potential due to the coronavirus (COVID-19) outbreak, the delayed Annual Budget Expenditure Act, and the drought. In this situation, there was an urgent need to coordinate financial and fiscal measures, as well as to support liquidity provision and debt restructuring for businesses and households. **Later on March 20, 2020, the Committee held a special meeting and voted unanimously to cut the policy rate by 0.25 percentage point from 1.00 to 0.75 percent, effective on March 23, 2020.** This was because the impact of the COVID-19 outbreak in the period ahead would be more severe than previously expected, which would severely affect the Thai economy. The outbreak also raised concerns among global financial markets and affected liquidity of Thai financial markets. **At the meeting on March 25, 2020, the Committee voted 4 to 2 to maintain the policy rate at 0.75 percent. Two members voted to cut the policy rate by 0.25 percentage point**, while one MPC member did not vote[1/]. The Committee viewed that the economic and inflation outlook remained consistent with the assessment at the special meeting. Most members would wait to assess the adjustment in the financial markets as well as additional reduction in interest rates by financial institutions. Most members deemed it appropriate to preserve policy space. Meanwhile, all members viewed that the government's announced relief measures and the loan repayment measures to assist borrowers affected by the COVID-19 were more targeted in addressing the problems. However, two members voted to cut the policy rate by 0.25 percentage point, as the extra accommodative monetary policy would in their view be commensurate with the significant economic contraction.

Looking ahead, the Committee would stand ready to use additional policy tools in an appropriate and timely manner. Such tools included the policy rate and other financial measures that would help enhance the policy rate transmission as well as loan extension by financial institutions to reach the target groups.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy was projected to slow down significantly as the COVID-19 pandemic severely affected tourism, manufacturing, and exporting sectors, as well as private consumption. The outbreak also led to excessive volatility in the global financial markets. Major advanced economies were expected to enter into a recession in 2020. **The Committee thus revised down economic growth projections for Thailand's trading partners to 0.2 percent in 2020 and 2.0 percent in 2021.** Nonetheless, **risks that trading partner economies would underperform the baseline projection increased.** Such risks included, for example, the COVID-19 outbreak which could become more severe and prolonged than expected, and financial stability risks which could deteriorate in many countries. On the other hand, there were possibilities that trading partners' economic growth would outperform the baseline projection owing to a faster-than-expected ending of the outbreak due to a successful development of vaccines or medicines, or larger-than-expected effects of financial and fiscal measures in many countries in stimulating the economy.

Governments and central banks in various countries worldwide promptly released measures to alleviate the impact of the COVID-19 outbreak. Such measures included greater government spending and extra accommodative monetary policy in conjunction with measures to safeguard financial stability and

[1/] The MPC member attended the meeting through video conference to consider the economic outlook, financial stability, and the appropriate monetary policy formulation but did not vote.

provide liquidity to financial markets. The Federal Reserve (Fed) held two unscheduled meetings in March 2020 and decided to cut the policy rate to a range of 0.00-0.25 percent and announced to purchase Treasury and agency mortgage-backed securities as much as needed to support the well-functioning of the financial markets. Meanwhile, the European Central Bank (ECB) and the Bank of Japan (BOJ) maintained negative policy rates and announced additional liquidity measures. Regional central banks eased their monetary policy further and also introduced measures to ensure stability of the financial markets following sustained capital outflows.

2. Financial Conditions and Financial Stability

Thai financial markets encountered volatilities similar to global financial markets. In March, investors' concerns over the COVID-19 pandemic increased, causing Thai government bond yields to rise rapidly across all tenors. **However, BOT measures helped improve stability of the financial markets. These measures included government bond purchases, the policy rate cut at the special meeting, and measures to stabilize the Thai financial markets.** As a result, government bond yields declined and overall financial conditions became more accommodative, partly reflected in the decreased lending rates.Meanwhile, overall corporate financing remained mostly unchanged. Financial institutions became more tightened in extending loans in line with economic conditions. Therefore, the BOT, together with related agencies, continued to implement measures to assist borrowers affected by the outbreak. These measures included both liquidity provision and debt restructuring. With regard to exchange rates, the baht against the U.S. dollar and the real effect exchange rate (REER) index depreciated considerably from the previous quarter.

The financial system faced greater risks from the short-term impact of the COVID-19 outbreak, although stability of financial institutions, external, and fiscal sectors, remained robust. Further risks that warranted monitoring were as follows. First, some investors rushed to redeem their investment units in daily fixed income funds, which were open for daily redemption, especially the funds investing in foreign assets. Some mutual funds therefore sold off their government bonds, thereby impacting liquidity and the functioning of the financial markets. Second, corporate bonds default risks increased in tandem with economic conditions, triggering knock-on effects to other related parties in the financial system, including cross-default and some saving cooperatives with corporate bond holdings. Third, ability to manage liquidity and debt servicing ability of households and businesses deteriorated. This could impact decisions to extend loans of financial institutions and in turn affect economic activities, resulting in a negative feedback loop. Fourth, oversupply in the real estate market was expected to rise due to falling demand for housing.

3. Economic and Inflation Outlook

The economy would contract significantly, expanding at a much lower rate than the assessment in the previous *Monetary Policy Report*. The Thai economy was expected to contract 5.3 percent in 2020 but would recover and expand 3.0 percent in 2021. The economy would remain at a level below its potential throughout the forecast horizon. This was mainly owing to the severe impact of the COVID-19 outbreak. In addition, farm income would be depressed by the drought, while public expenditure would experience a larger impact than previously assessed due to the delayed enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020). Nevertheless, government's implemented measures would be the key mechanism to alleviate the impact on the economy despite being unable to offset a significant contraction in aggregate demand.

Assumptions underlying the economic projection in this *Report* include (1) the COVID-19 outbreak in Thailand would be contained within the second quarter of 2020 according to the Ministry of Public Health estimate, and (2) the projection took into account financial and fiscal measures implemented before March 25, 2020, not including any measures thereafter.

The value of Thai merchandise exports in 2020 would markedly shrink both in terms of volume and prices. Export volume would be affected by the COVID-19 outbreak, which resulted in a significant demand slowdown of trading partner economies and supply chain disruptions. Meanwhile, export prices would significantly decline in line with crude oil prices. However, the value of merchandise exports was expected to expand slightly in 2021 once the spread of COVID-19 subsided. Nevertheless, **the Committee assessed that the value of merchandise exports would contract 8.8 percent in 2020, before expanding slightly at 0.2 percent in 2021.**

Exports of services was expected to severely contract as the COVID-19 outbreak led to travel restrictions that affected foreign tourist figures of all nationalities. Moreover, there would also be an impact on tourist spending per head. However, the number of foreign tourists was expected to gradually improve in the second half of 2020. Nevertheless, tourists would start to travel once the vaccines for the coronavirus were to be widely distributed. **The number of foreign tourists was projected to be 15.0 and 20.0 million in 2020 and 2021, respectively.**

The current account balance was projected to register a surplus of 19.4 and 19.2 billion U.S. dollars in 2020 and 2021, respectively. The services balance was revised from being in surplus to deficit, as tourist receipts declined considerably in line with the decline in the number of tourists and spending per head. Meanwhile, the trade balance was expected to record a larger surplus due to a large contraction in imports value. Import volume contracted in line with imports of raw materials, intermediate and capital goods, while import prices declined in tandem with global crude oil prices.

Private consumption was expected to contract in 2020, which was the first contraction since 2009. Suspension of a wide range of economic activities affected consumption in the services and away-from-home spending categories. Meanwhile, farm income was expected to decline due to a more-severe-than-expected impact of the drought. However, government measures to alleviate the impact of the COVID-19 outbreak would partly help relieve the impact on private consumption to a certain level.

Private investment was projected to contract in 2020 in line with both domestic and external demand. **However, private investment was expected to improve in 2021** thanks to a gradual recovery in exports, the PPP projects in infrastructure investment projects that had been postponed, as well as a greater clarity pertaining to investment in the 5G network following the auction in the first quarter of 2020.

Public expenditure, both consumption and investment, was expected to expand and would be a key driver of the Thai economy. However, although government agencies adjusted the procurement process for a faster disbursement, this could not offset the impact of the delayed enactment of the Annual Budget Expenditure Act in the previous period. SOE investment was revised down in line with project postponement and a lower fiscal budget on transportation projects. Meanwhile, the first and second packages of government measures to mitigate the impact of the COVID-19 outbreak, totaling 464 billion baht or 2.8 percent of GDP (excluding tax relief measures worth approximately 260 billion baht or 1.5 percent of GDP), were mostly measures on credit and transfers. This was expected to directly cushion private consumption and investment, not through public consumption and investment.

Headline inflation would be negative due to the sharp decline in crude oil and energy prices. Core inflation was expected to be negative as well in line with a large decrease in demand-pull inflationary pressures. Headline inflation was projected to be negative at -1.0 percent in 2020, before gradually increasing to 0.3 percent in 2021 consistent with the gradual economic recovery outlook. However, headline inflation would remain below the inflation target.

Economic and inflation projections were highly uncertain and mainly depended on the situation of the COVID-19 spread and the success of the containment measures. Risks to the projections tilted downward. The possibilities that the Thai economy would underperform the baseline projection would be due to the following factors. First, the global economy could enter into a recession should the

outbreak become more severe and prolonged than expected, exerting additional adverse impacts on Thailand's merchandise exports. Second, global financial stability risks might affect the Thai economy in the medium term. Third, Thailand's tourism might not recover in the second half of 2020 if the outbreak in Thailand and other countries could not be contained within the second quarter of 2020. Fourth, the government might need to implement stricter measures to control the outbreak, which might cause some businesses to shut down operations or significantly reduce employment. Fifth, more-severe-than-expected impact of the drought could affect farm income. However, **there were possibilities that the Thai economy would outperform the baseline projection** due to (1) a faster-than-expected development of vaccines and medicines for the COVID-19, (2) better-than-expected impacts of additional government's stimulus measures and measures to assist borrowers, and (3) a higher-than-expected domestic demand as a result of public infrastructure investment and PPP projects, which might incentivize the private sector to invest more and faster. Moreover, **there remained downside risks to headline and core inflation projections** in line with risks to crude oil prices and the economic growth projection.

1. Global Economy

The COVID-19 pandemic significantly weighed on the outlook of Thailand's trading partner economies, particularly major advanced economies that were expected to enter into a recession in 2020. However, financial and fiscal measures would help alleviate the impact and shore up the economy in the period ahead.

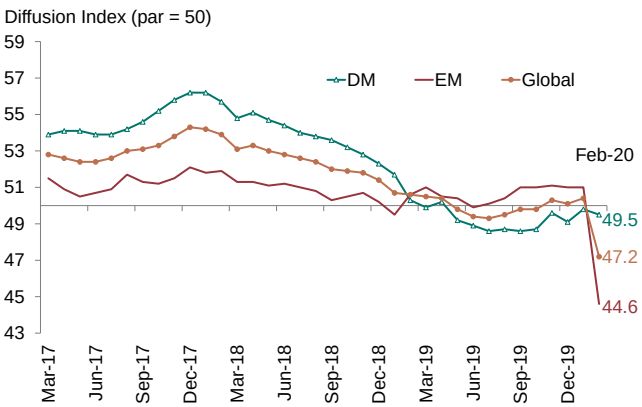
In the fourth quarter of 2019, the U.S. economy expanded with growth rate largely unchanged from the previous quarter. Exports improved, while private consumption could still grow on account of robust labor markets and accommodative monetary conditions. Meanwhile, private investment continued to contract. **The euro area economy** expanded at a slower pace due to country-specific factors in certain countries. For example, the protest in France against the pension reforms toward the end of the year led private consumption to decelerate. **The Japanese economy** contracted, as private consumption was affected by an increase in consumption tax in October, while investment and manufacturing sector also contracted as a result of Typhoon Hagibis.

In 2020, major advanced economies were projected to continue to contract and enter into a recession as a result of the COVID-19 pandemic. Private consumption and tourism-related businesses were severely affected as the U.S. imposed travel restrictions on European countries. Several European countries announced border closures and lockdown of cities, as well as advised its citizens against all but essential travel. Japan also announced a postponement of the Summer Olympic Games. Meanwhile, production and exports in certain industries, such as automotive and electronics, encountered supply chain disruptions, while the U.S. energy sector was affected by a significant fall in oil prices. Nevertheless, financial and fiscal measures which were already implemented and forthcoming would help alleviate the impact and shore up the economies of major advanced countries in the period ahead.

The Chinese and Asian economies expanded at a slower pace due to the COVID-19 outbreak. However, financial and fiscal measures would help alleviate the impact and support the economic expansion to some extent.

The Chinese economy in the fourth quarter of 2019 expanded at a pace similar to that of the previous quarter. Exports and investment improved, while private consumption slowed down slightly. **In 2020, the Chinese economy was projected to slow down significantly** as a result of the COVID-19 outbreak, which started to intensify since the Chinese New Year. The Chinese authorities, therefore, announced a lockdown of several major cities, particularly in Hubei province, and extended holiday periods. This resulted in a severe impact on economic activities, especially in the manufacturing sector.

Chart 1.1 Global manufacturing was expected to decline significantly in both major advanced and emerging economies.
Global manufacturing Purchasing Manager Index (PMI)

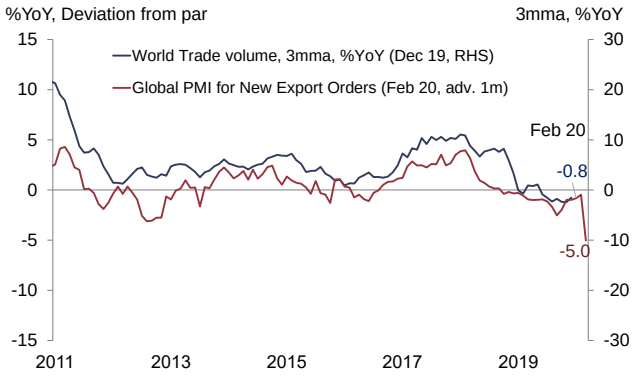


Note: DM = Developed Markets; EM = Emerging Markets
Source: CEIC, Markit PMI, JP Morgan (Global PMI)

(Chart 1.1). Although the number of new infections in China fell significantly and the authorities allowed some businesses to resume operations after the extended holidays, quarantine measures and travel restrictions between cities that remained in place prevented businesses from performing full operations. Moreover, since there were still infected patients travelling into China, it would take some time before the situation returned to normal. Nevertheless, fiscal measures which were continuously introduced would in part help to alleviate the impact and prevent the Chinese economy from entering into a recession.

In the fourth quarter of 2019, Asian economies (excluding Japan and China) expanded at a slower pace than the previous quarter. This was because production and exports continued to be affected by trade tensions, as reflected in a slower-than-expected recovery in global trade volume (Chart 1.2). Moreover, prolonged protests in Hong Kong resulted in continued contraction of the economy. **In 2020, Asian economies were projected to slow down significantly from the previous assessment**, as the COVID-19 spread to many countries such as South Korea, Hong Kong, Malaysia, and the Philippines. Moreover, nationwide or citywide lockdown measures imposed by many countries weighed on private consumption and tourism sector. Meanwhile, the manufacturing and export sectors were affected by supply chain disruptions from linkages with China's production chain, and slower-than-expected recovery of the electronics cycle. Nevertheless, governments in many countries had introduced stimulus measures which should help support the recovery of Asian economies in the period ahead.

Chart 1.2 Global PMI for New Export Orders reflected a much slower-than-expected recovery in global trade volume



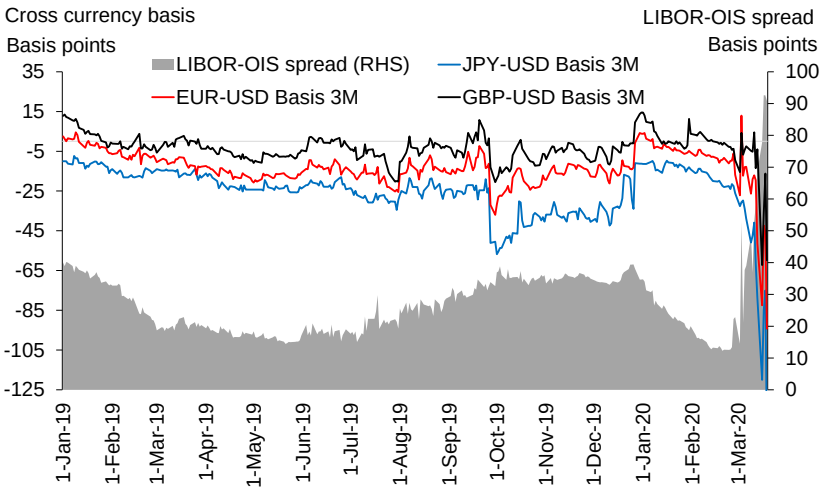
Source: CPB, JP Morgan

The COVID-19 outbreak and a sharp fall in oil prices contributed to excessive volatility in the global financial markets, both in major advanced economies and emerging markets.

In the first quarter of 2020, concerns in the global financial markets heightened, triggering risk-off sentiment, as a result of the COVID-19 outbreak and a marked decline in oil prices. This was reflected in a sharp increase in the VIX Index[2/] since late February. Investors thus continued to sell off their risky assets. This was particularly the case in major advanced economies, where massive equity sell-offs led to sharp plummets of stock market indices and triggered circuit breakers in some countries. As a result, long-term U.S. treasury yields were volatile, reaching historical low levels at the beginning of March. This was due to increased demand for U.S. treasuries following a fall in prices of risky assets. However, U.S. treasury yields rose rapidly in the middle of March, as investors cut their holdings of all types of assets, including safe-haven assets, in exchange for U.S. dollar cash. The sharp surge in demand for U.S. dollars in the global financial markets resulted in a dollar liquidity squeeze, as reflected in a more negative cross currency basis (Chart 1.3). Meanwhile, U.S. corporate bond yields rose markedly due to concerns over debt servicing ability of affected businesses.

[2/] VIX Index is a measure of the stock market volatility implied by S&P 500 Index options.

Chart 1.3 Sharp surge in demand for U.S. dollars of investors amid uncertainties in the global financial markets resulted in a dollar liquidity squeeze.



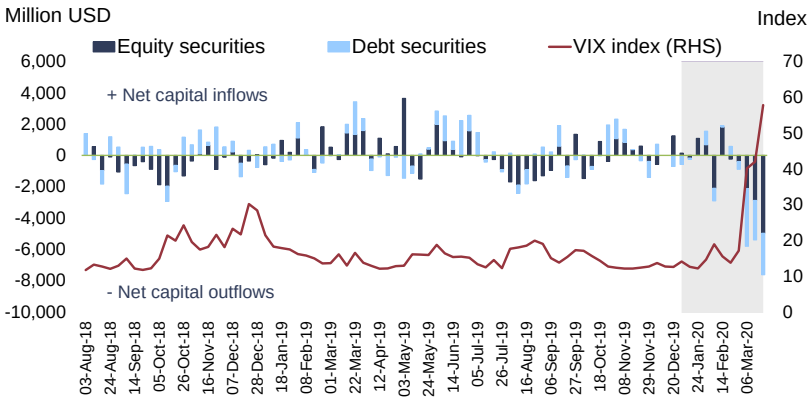
Note: a more negative cross currency basis reflected increased demand for holding U.S. dollar cash relative to other currencies. A surge in LIBOR-OIS spread indicated higher term premium due to investors' views of higher credit risks in lending markets.

Source: Bloomberg, compiled by the BOT (data as of 19 March 2020)

In emerging markets, there were signs of capital outflows since when the outbreak was still contained in China and Asia. Outflows, from both equity and bond markets, then increased since late February (Chart 1.4), causing the stock market index to fall sharply and government bond yields to rise. Overall global financial conditions tightened significantly. Going forward, the global financial markets would remain highly volatile due to short-term risks. These included uncertainties regarding the impact of COVID-19 on the economy and financial stability, as well as competition on prices and production volume among major oil producers, which would weigh on the outlook of both oil and commodity prices in the period ahead.

Chart 1.4 EMs experienced capital outflows from both equity and bond markets following rising concerns among investors as a result of the COVID-19 outbreak and a marked decline in oil prices.

Net capital inflows to EMs* and VIX index (weekly)



Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey

Sources: Bloomberg and Institutional Institute of Finance

Governments worldwide introduced financial and fiscal measures in order to alleviate the economic impact of COVID-19 and curb volatilities in the financial markets.

Governments in various countries worldwide promptly released measures to mitigate the impact of the economic slowdown. In major advanced economies and emerging markets, governments announced a series of measures to shore up the economy, as well as introduced targeted measures. This particularly included measures to support businesses and SMEs, and measures to support households and employment. Meanwhile, **advanced economies' central banks eased their monetary policy further.** The Federal Reserve (Fed) cut its policy rate from 1.50-1.75 percent to 0.00-0.25 percent in two unscheduled meetings in March 2020. Meanwhile, the European Central Bank (ECB) kept its policy rate at a low level and injected liquidity in the financial system through longer-term refinancing operations (LTROs). Moreover, the ECB undertook targeted longer-term refinancing operations (TLTROs) in order to support credit extension by financial institutions to the real economy. The Bank of Japan (BOJ) maintained its policy rate at a low level, expanded its asset purchases in the equity market, and introduced Special Funds-Supplying Operations in order to facilitate greater access to credits for businesses. **Regional central banks also eased their monetary policy further** by cutting the policy rates in order to help mitigate the economic impact of the COVID-19 outbreak. Some central banks introduced measures to reduce financial costs for borrowers, such as temporary debt moratorium measures, and a low-interest loan program for financial institutions to extend credits to the private sector.

In addition, many countries simultaneously introduced measures to safeguard stability of the financial markets stemming from tightening liquidity conditions in the global financial markets. The Fed injected liquidity in the bond market through the purchases of treasuries and agency mortgage-backed securities in the amounts as much as needed to support the well–functioning of the financial markets. Moreover, the Fed established facilities to support the issuances of bonds in the primary and secondary markets[3/], and also injected liquidity into the financial institutions' system via the establishments of various funds. This would ensure sustained flows of credit from financial institutions to businesses and households[4/]. Meanwhile, the ECB increased the size of its purchases of government and corporate bonds via the Pandemic Emergency Purchase Program (PEPP) worth 750 billion euros. The program was made more flexible than the previous one, for instance, by allowing extensive holdings of bonds issued by any single member country. **Regional central banks announced measures to ensure stability of the financial markets following sustained capital outflows.** These included increasing the purchases of government bonds, as well as providing liquidity via other instruments such as FX swap transactions, expanding the set of

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[3/] The Fed established the Primary Market Corporate Credit Facility (PMCCF) and the Secondary Market Corporate Credit Facility (SMCCF) to support credit extension to large corporates. The facilities would purchase newly issued investment grade corporate bonds in the primary market, and purchase debt securities and exchange-traded funds (ETFs) in the secondary market.

[4/] The liquidity facilities included following examples. (1) Under the Primary Dealer Credit Facility (PDCF), the Fed would extend credit to its primary dealers. The loan would be fully collateralized with maturities up to 90 days. (2) Under the Money Market Liquidity Facility (MMLF), the Fed would provide liquidity, with maximum maturities of 12 months, to financial institutions to purchase assets from primary money market fund according to conditions under the MMLF. (3) Under the Term Asset-Backed Securities Loan Facility (TALF), the Fed would extend loans fully collateralized by eligible asset-backed security (ABS) to finance student, auto, and credit card loans. This would increase flows of credit to households and businesses.

eligible collaterals for open market operations, and cutting reserve requirement ratio (RRR). Meanwhile, some countries announced temporary closures of their financial markets[5/].

Growth projections for Thailand's trading partner economies in 2020 were revised down significantly due to impact of the COVID-19. In 2021, however, trading partner economies were expected to recover once the spread gradually subsided. Nonetheless, risks that trading partner economies would underperform the baseline projection increased, as the COVID-19 pandemic could become more severe and prolonged than expected.

Trading partner economies in 2020 were expected to expand at a much slower pace than previously assessed in the previous *Monetary Policy Report*. This was because global COVID-19 pandemic severely affected manufacturing, consumption, and investment. Moreover, the services sector was significantly disrupted and was expected to recover more slowly than the manufacturing sector. Nevertheless, private consumption would be somewhat supported by falling oil prices. At the same time, financial and fiscal measures that were already implemented and forthcoming would partly help shore up the economy. In 2021, the COVID-19 outbreak was expected to gradually subside. Consequently, trading partner economies would somewhat recover. **The Committee thus revised down economic growth projections for Thailand's trading partners to 0.2 percent in 2020 and 2.0 percent in 2021** (Table 1.1).

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2019*	2020	2021
United States	18.2	2.3	-0.3 (1.8)	1.4
Euro area	9.9	1.2	-2.6 (1.1)	-0.3
Japan	14.3	0.7	-1.7 (0.4)	0.7
China	17.0	6.1	3.9 (5.9)	5.5
Asia (excluding Japan and China)**	32.4	2.7	0.2 (3.2)	2.1
Total***	100	2.7	0.2 (2.7)	2.0

Note: *Outturn

**Weighted by 7 trading partners' shares in Thailand's exports in 2019, namely, Singapore (5.1%), Hong Kong (6.8%), Malaysia (6.1%), Taiwan (2.3%), Indonesia (5.3%), South Korea (2.7%), and the Philippines (4.0%)

***Weighted by 13 trading partners' shares in Thailand's exports in 2019 (including the United Kingdom and Australia)

( ) as reported in Monetary Policy Report, December 2019

[5/] The Philippines announced closures of the stock, bond, and currency markets during March 17-18, 2020 in an attempt to cope with the volatility and the speculation in the financial markets following the COVID-19 outbreak.

The Committee assessed that there were higher possibilities that trading partner economies would underperform the baseline projection, as the COVID-19 outbreak could become more severe and prolonged than expected. Moreover, there were other risk factors, such as (1) financial stability risks which could arise in many countries as a result of high volatility and tightening liquidity in the global financial markets, particularly risks of a sudden stop in emerging market economies, (2) geopolitical risks which could escalate and weigh on the commodity market, (3) trade tensions which could intensify once again, (4) uncertainties regarding trade deal between the European Union and the United Kingdom, and (5) wealth effect arising from volatile financial markets and a sharp fall in asset prices**. On the other hand, there were possibilities that trading partners' economic growth would outperform the baseline projection** owing to (1) a faster-than-expected ending of the outbreak due to effective healthcare measures in many countries and successful development of vaccines and medicines for the COVID-19, and (2) larger-than-expected effects of financial and fiscal measures in many countries in stimulating the economy.

Dubai crude oil prices in the first quarter of 2020 declined markedly from the previous quarter. This was in line with the global economic slowdown as a result of the COVID-19 outbreak and the announcement on oil price reduction by Saudi Arabia after the OPEC and Russia failed to reach an agreement on a crude oil production cut.

Dubai crude oil prices in the first quarter of 2020 fell sharply from the previous quarter due to the following factors. First, the COVID-19 outbreak resulted in a lower demand for oil in line with the global economic conditions, particularly fuels used in transportation and travel. **Second, Saudi Arabia announced a reduction on crude oil prices after Russia failed to agree to the OPEC's proposal** to additionally cut oil production at the meeting on March 6, 2020. Saudi Arabia also intended to gain market shares from other higher-cost oil producers such as the U.S. Moreover, Saudi Arabia and Russia announced an increase in their crude oil productions by 2.5 million and 0.5 million barrels per day respectively, starting in April 2020. As a result, Dubai crude oil prices fell substantially from their average of 54.2 U.S. dollars per barrel in February 2020 to below 30.0 U.S. dollars per barrel since the middle of March 2020 (Chart 1.5).

Due to factors mentioned above, together with uncertainties surrounding the COVID-19 outbreak situation that could potentially impact global oil demand significantly, **the Committee thus revised down the projection for Dubai crude oil prices in 2020** from 62.5 U.S. dollars per barrel to 35.0 U.S. dollars per barrel. In 2021, Dubai crude oil prices were expected to remain low at 34.0 U.S. dollars per barrel. Crude oil prices would gradually increase in line with the global economic recovery and a reduction in oil supply from high-cost producers such as the U.S. Meanwhile, **risks to the projection of Dubai crude oil prices were expected to tilt downward,** as the COVID-19 outbreak could become more prolonged and affect the global economy more than expected. Nevertheless, Dubai oil prices could increase somewhat as a result of geopolitical factors and the possibility of the OPEC and Russia reaching a production cut agreement.

Chart 1.5 Dubai crude oil prices in the first quarter of 2020 fell sharply from the previous quarter due to slowing global economy as a result of the COVID-19 outbreak and Saudi Arabia's announcement to cut crude oil prices after the OPEC and Russia failed to agree on crude oil production cut

Crude oil price projection



2. The Thai Economy

2.1 Recent Developments

The Thai economy in the fourth quarter of 2019 expanded at a slower pace due to both domestic and external demand.

In the fourth quarter of 2019, the Thai economy expanded 1.6 percent from the same period last year. The growth rate was lower than the assessment in the previous *Monetary Policy Report*, mainly due to domestic and external demand slowdown. The prolonged contraction in Thai merchandise exports prompted firms to reduce the number of workers and working hours. Firms also adjusted their employment pattern from monthly to daily contract. Household income and consumer confidence, that continued to weaken, caused private consumption to moderate. The government thus introduced short-term economic stimulus measures toward the end of 2019, which helped shore up purchasing power to some extent. However, the government was unable to lend sufficient support to the economy on account of the delayed enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020). Therefore, public expenditures contracted in both current and capital expenditures. Tourism growth moderated as a result of both the waning low base effect from the Phuket tour boat sinking incident and lower tourist spending in line with slower economic growth of trading partners. Moreover, weaker economic activities resulted in the continued slowdown of private investment.

The Thai economy experienced a more severe impact of the COVID-19 outbreak.

In the first quarter of 2020, the Thai economy faced a more severe impact of the COVID-19 outbreak, which started to intensify since late January. Following the Chinese government's prohibition of the Chinese outbound travel and the rapid spread of the virus to other countries, the number of foreign tourist arrivals sharply contracted in almost all nationalities. As a result, tourism and related businesses were severely affected. Moreover, the COVID-19 outbreak triggered supply chain disruptions and closure of transport routes in some countries, further affecting Thai merchandise exports (excluding gold). Nonetheless, some Thai export products benefited from export substitution of Chinese goods in the U.S. market as well as exports of medical-related products and other basic necessities during the COVID-19 outbreak. However, this was not sufficient to offset the negative impact of weaker demand from trading partners.

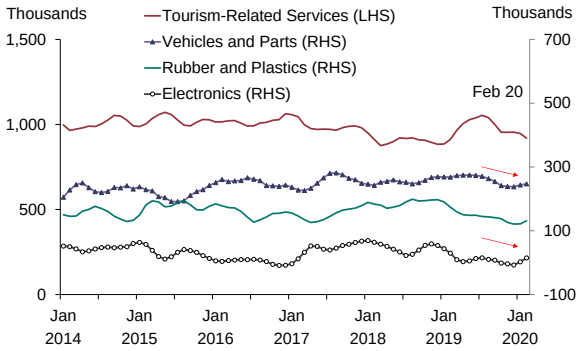
In addition, the COVID-19 outbreak resulted in broader economic impacts, especially for SMEs that were short of liquidity and had to shut down operations. As more workers lost their jobs and consumer confidence dropped to its lowest level in 19 years, private consumption indicators thus pointed to a slowdown. In particular, consumer spending in services category contracted considerably, where only spending on necessities that could still expand. Private investment indicators further contracted, driven by substantially weaker domestic and external demand and problem in transporting imported capital goods from China. Despite the introduction of a series of measures to alleviate the impact on those affected by the outbreak, the government was unable to provide a sufficient support to the economy in this quarter as problem regarding parliamentary voting process to approve the Annual Budget Expenditure Act caused further delay in the enactment.

Overall household purchasing power significantly weakened. The COVID-19 outbreak further affected employment and non-farm income, while the drought might cause a more-severe-than-expected impact on farm income.

The COVID-19 outbreak further weakened an already-fragile household purchasing power. **Non-farm households** faced greater downside risks due to changes in employment pattern from monthly to daily contract (Chart 2.1), temporary or partial suspension of business operations (Chart 2.2), and a reduction in overtime (OT) working hours. These downside risks, which had initially been concentrated in export-related manufacturing sectors, spread to tourism-related services sector, such as hotel and restaurants and retail trade. As a result, non-farm income declined sharply.[6/]

Chart 2.1 Businesses in tourism-related services and export-related manufacturing sectors changed employment patterns substituting monthly to daily employment contract

Differences between the number of monthly- and daily-paid workers

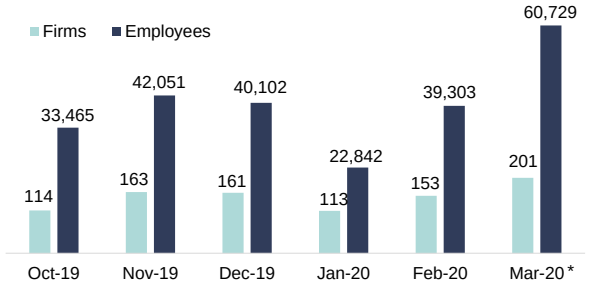


Note: The decrease in the difference between monthly- and daily-paid workers implies a higher number of daily employment contract

Source: National Statistics Office

Chart 2.2 Non-farm income fell sharply as the number of employees and firms affected from the temporary suspension of business operation (section 75) was expected to rise

Temporary suspension of business operation (Section 75)



Note: Section 75 figures are the sum of firms reported to the Ministry of Labor in each month for the temporary suspension of business operation.

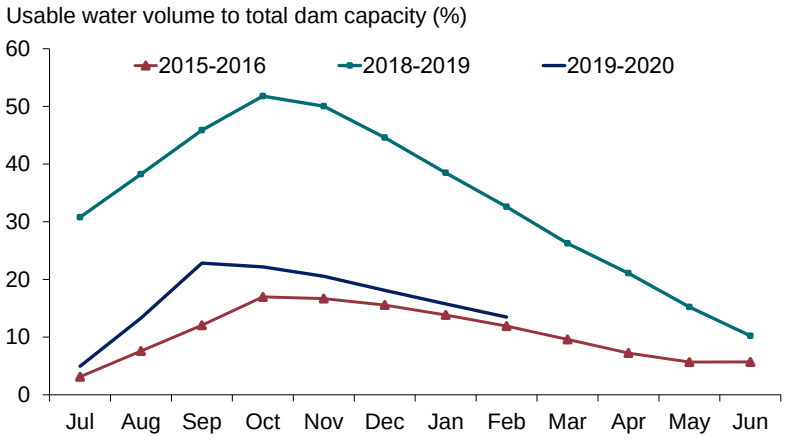
*Data as of 23 March 2020

Source: Department of Labor Protection and Welfare, Ministry of Labor

Farm income in 2020 was projected to contract more than previously assessed, mainly due to a significantly lower agricultural output, especially off-season rice and sugarcane, as a result of the drought. Although prices of these agricultural products would rise somewhat due to the drought, the price increase would not offset the impact of such large fall in output. Moreover, the drought situation could be more severe than expected because the level of usable dam water (including dams in the northern and central regions) was at a similarly critical level as in 2015-2016 (Chart 2.3), while risk of El Nino increased. Should the rainy season in Thailand delay, the usable dam water level would not be sufficient to support farming in the latter half of 2020. This would further affect output, especially in-season rice and other crops including cassava, vegetable, and fruits.

[6/] The COVID-19 severely affected the labor market because over half of tourism-related workers were self-employed with unstable income. Moreover, most workers were employed in jobs where "work from home" was not possible. For more information, please see the link Thai labor market conditions and outlook (March 2020).

Chart 2.3 Usable dam water was at a similarly critical level as in 2015-2016

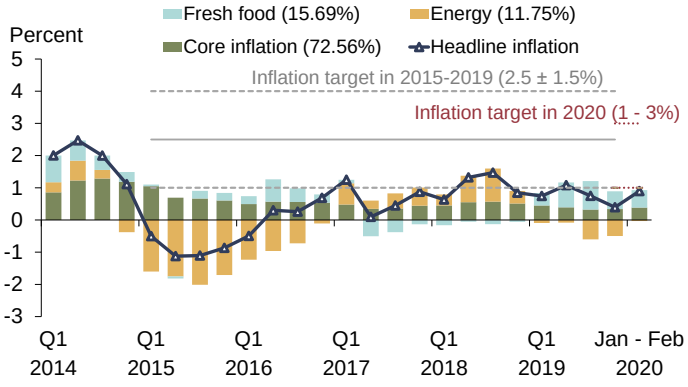


Source: Royal Irrigation Department, calculations by Bank of Thailand
Note: usable water volume to total dam capacity at end month

Headline inflation in January-February increased due to the base effect of energy prices, but was projected to decline in March in line with global crude oil prices.

Chart 2.4 Headline inflation in Jan – Feb increased from the previous quarter mainly due to the base effect of energy prices

Headline inflation and inflation target



Note: Headline inflation for the first 2 months of 2020 recorded 0.90 percent
( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculations by Bank of Thailand

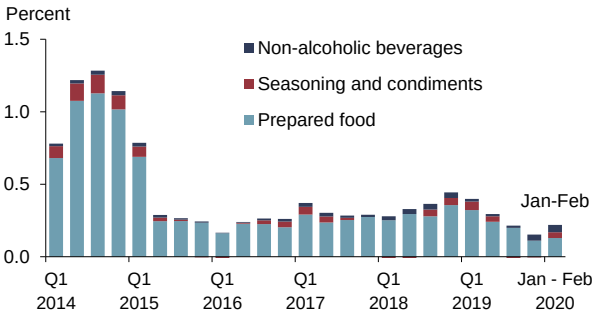
Headline inflation in the first two months of the first quarter of 2020 averaged at 0.90 percent, up from 0.40 percent in the previous quarter (Chart 2.4). The increase was in tandem with the smaller decline in energy prices, as domestic retail oil prices decreased from the same period last year following the decision to reduce contributions to the Oil Fund. Fresh food inflation was largely unchanged from the previous quarter. Fresh food prices in several product categories decreased, such as vegetables and fruits, eggs and dairy products, as well as rice, flour, and flour products. However, the decrease was in part offset by higher meat prices, mainly in line with pork prices given lower pork output. Headline inflation in March was projected to fall in line with global crude oil prices.

Core inflation in the first two months of the first quarter of 2020 averaged at 0.53 percent, up from 0.47 percent in the previous quarter. Food-in-core inflation increased (Chart 2.5), driven by higher prices of seasonings and condiments. This was partly owing to an increase in vegetable oil prices on account of higher palm costs. Core inflation in the non-food categories declined slightly in line with prices of personal care products (Chart 2.6). This was partly because firms rolled out more promotional activities. In other categories, prices slightly increased due to weaker demand in line with economic conditions (Table 2.1).

Short-term (one-year ahead) inflation expectations according to the survey of professional forecasters in March 2020 recorded 1.1 percent, a slight decline from 1.2 percent in the previous quarter. Inflation expectations based on the survey of businesses in February 2020 stood at 1.6 percent, down slightly from 1.7 percent in the previous quarter. **Long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in October 2019 remained at 2.0 percent. In addition, long-term inflation expectations based on a macro-finance term structure model with bond yields and macroeconomic data stabilized at 1.6 percent in the fourth quarter of 2019.

Chart 2.5 Core inflation in the food category (28% of core inflation) increased mainly on account of a rise in seasoning and condiments prices

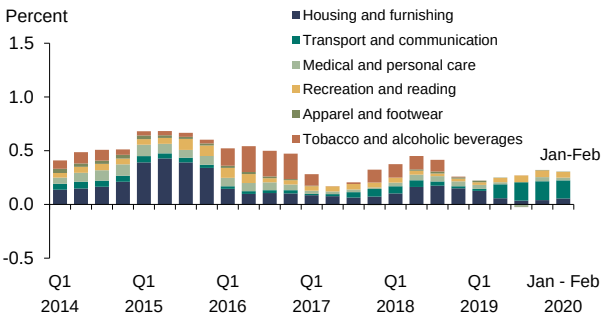
Contribution* to core inflation in the food category



Note: *Contributions to core inflation decompose core inflation into inflation of each component within the basket, weighted by its corresponding share.
Source: Ministry of Commerce, calculations by Bank of Thailand

Chart 2.6 Core inflation in the non-food category (72% of core inflation) slightly declined mainly in line with personal care prices

Contribution* to core inflation in the non-food category



Note: * Contributions to core inflation decompose core inflation into inflation of each component within the basket, weighted by its corresponding share.
Source: Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

Annual percentage change	2018				2019				2020
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Jan-Feb
Headline Consumer Price Index (Headline CPI)	0.64	1.31	1.47	0.84	0.74	1.08	0.61	0.40	0.90
Core Consumer Price Index (Core CPI)	0.61	0.76	0.78	0.71	0.62	0.54	0.45	0.47	0.53
Fresh food	-1.04	-0.35	-0.82	-0.35	2.51	4.98	5.79	3.58	3.47
Energy	3.01	7.30	9.11	3.39	-0.78	-0.67	-4.95	-4.09	-0.26

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

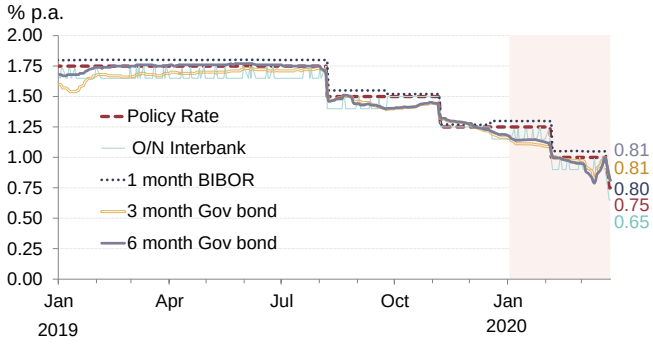
Short-term government bond yields and commercial banks' interest rates fell in line with the policy rate. Long-term government bond yields also declined in tandem with U.S. treasury yields, before rising rapidly towards the end of the quarter on the back of concerns over the COVID-19 situation.

During January-February 2020, **short-term money market rates and short-term government bond yields** fell from the previous quarter in line with the policy rate cut on February 5, 2020 (Chart 2.7). **Medium-term and long-term government bond yields** (Chart 2.8) decreased in tandem with U.S. treasury yields, as investors viewed that the COVID-19 began to become widespread from originally contained within China, which could significantly affect the global economy. In addition, two-year Thai government bond yields were subjected to additional pressures from increased demand from investors. This was on account of the Ministry of Finance's bond switching transactions in which two-year government bonds were set as the source bond. Meanwhile, **overall corporate bond yields** decreased in line with government bond yields.

However, in March 2020, **Thai government bond yields increased rapidly across all tenors** after the World Health Organization (WHO) declared COVID-19 as a pandemic. Foreign investors quickly sold off Thai government bonds in exchange for cash in U.S. dollars. Meanwhile, some domestic investors rushed to redeem their investment units in daily fixed income funds, which allowed for daily redemption, especially the funds investing in foreign assets. Some mutual funds thus sold off quality assets before their maturities, particularly Thai government bonds, in order to have enough cash to return to unitholders. Consequently, Thai government bond yields rose quickly in some tenors, putting a strain on the liquidity and functioning of the financial markets. This was reflected in the abnormally high bid-ask spreads. However, Thai government bond yields decreased slightly, thanks to the Bank of Thailand's government bond purchase program to provide liquidity in the bond markets, the policy rate cut at the special meeting of the MPC on March 20, 2020, and the joint announcement of measures to stabilize the Thai financial markets in corporation with related agencies. As a result, mutual funds slowed their sell-off of government bonds to some extent *(**Box**: Measures to stabilize the Thai financial markets* in chapter 3*)*.

Chart 2.7 Short-term government bond yields declined in line with the policy rate but temporarily increased toward the end of the quarter driven by investors' concerns over the COVID-19 situation

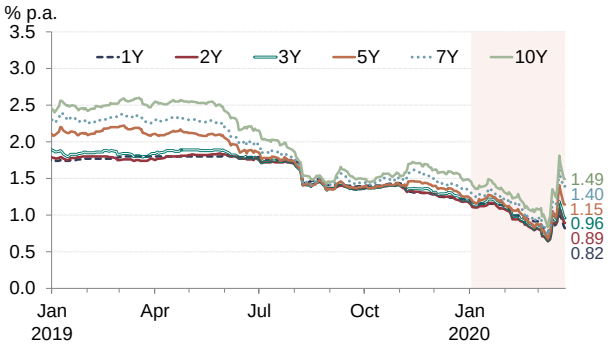
Short-term rates in financial markets



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 24 March 2020)

Chart 2.8 Government bond yields continued to decline, before temporarily rising toward the end of the quarter due to investors' accelerated sell-off from concerns over COVID-19

Government bond yields

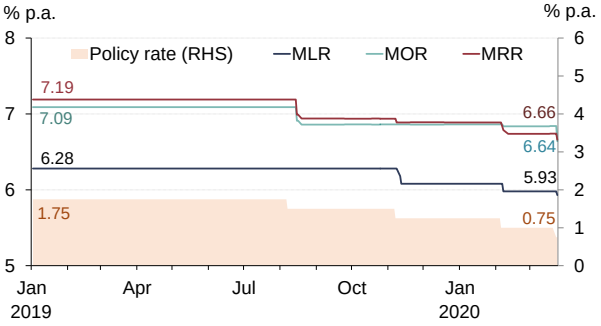


Source: Thai Bond Market Association (Thai BMA) (data as of 24 March 2020)

The five largest commercial banks reduced lending rates in all categories in February and March 2020. The average minimum loan rates (MLR) decreased to 5.93 percent, the average minimum retail rates (MRR) slightly declined to 6.66 percent, and the average minimum overdraft rates (MOR) fell to 6.64 percent (Chart 2.9). Meanwhile, the new loan rates (NLR)[7/] in January 2020 remained low at 3.83 percent (Chart 2.10). **Deposit rates** of the top five commercial banks continued to decline in tandem with lending rates, especially for 3-month, 6-month and 12-month fixed deposits as well as savings rates (Chart 2.11).

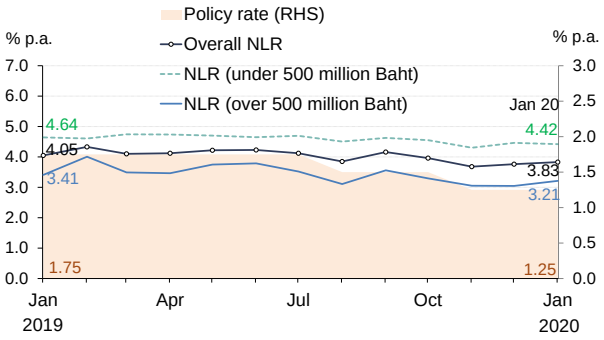
[7/] NLR is the weighted average of interest rates on new loan contracts extended by 14 Thai commercial banks each month. The loan contracts exclude consumer loans, credit card loans, repurchase agreements, bank guarantees, as well as loans extended to financial intermediaries, the public sector and non-residents. The dataset covers loans with value of 20 million baht or higher for all loan types, purposes and maturities, and includes both secured and unsecured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

Chart 2.9 MLR, MOR, and MRR of commercial banks declined in tandem with the policy rate



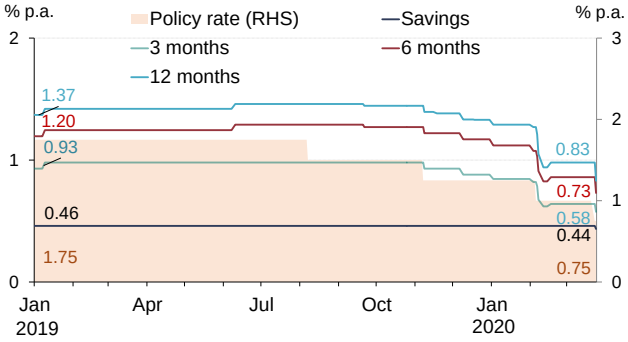
Note: Averages of 5 largest commercial banks' lending rates
Source: Bank of Thailand (data as of 24 March 2020)

Chart 2.10 Overall New Loan Rate (NLR) of commercial banks slightly increased but remained low



Source: Bank of Thailand (data as of 24 March 2020)

Chart 2.11 Commercial banks gradually reduced deposit rates of 3, 6, and 12 months fixed deposits and savings



Note: Averages of 5 largest commercial banks' lending rates
Source: Bank of Thailand (data as of 24 March 2020)

Private sector financing remained mostly unchanged from the end of the fourth quarter of 2019. Business credit expanded at a slightly faster pace, while household credit slowed down slightly.

Private credit[8/] expanded 3.4 percent in January 2020 (Chart 2.12), largely unchanged from December 2019, after having decelerated in 2019 in line with economic conditions. **Business credits expanded slightly** by 2.2 percent in January 2020, as business loans contracted slightly by 0.6 percent while corporate bond issuance grew at a relatively steady pace of 13.1 percent (Chart 2.13). Real estate and manufacturing businesses continued to raise funds through bond issuance, mainly for working capital and to repay their matured debt. **Household credit growth slowed down slightly** to 4.5 percent, largely due to slower growth in auto-leasing.

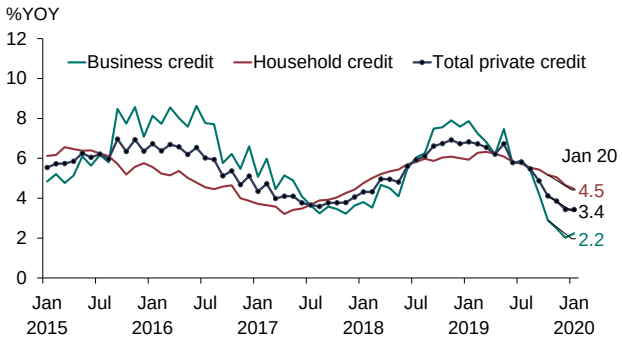
In the first quarter of 2020, demand for private sector financing rose, especially among businesses affected by the COVID-19 outbreak. Nevertheless, based on the survey in January 2020, financial institutions were expected to tighten credit underwriting standards in the first

[8/] Loans extended to the private sector consist of those extended by (1) other depository corporations (ODCs) including commercial banks, specialized financial institutions, finance companies, saving cooperatives and market mutual funds (2) other financial institutions (OFCs) and (3) non-residents.

quarter of 2020[9/], especially for business loans extended to large businesses and SMEs, as well as credit card loans, mainly given concerns over economic conditions.

Chart 2.12 Private credit remained largely unchanged as household credit slowed down slightly while business credit slightly expanded

Growth of private credit

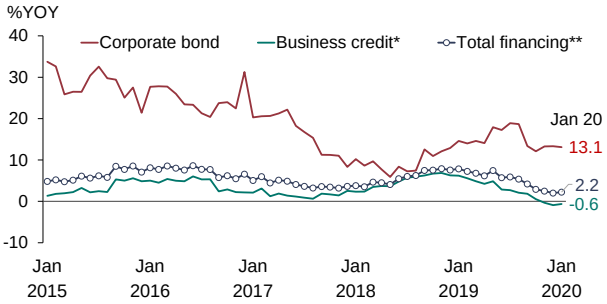


Note: 1) Private credits include credit extended by (1) Other Depository Corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds, (2) Other Financial Corporations (OFCs), and (3) non-resident
2) The definition of private credit was redefined to cover broader types of lenders and financial instruments. The data was first published in the MPR in 2019Q4, where the data was available from January 2012.

Source: Bank of Thailand

Chart 2.13 Private sector financing accelerated slightly mainly on account of a smaller contraction in credit growth

Growth of business credit and corporate bond outstanding



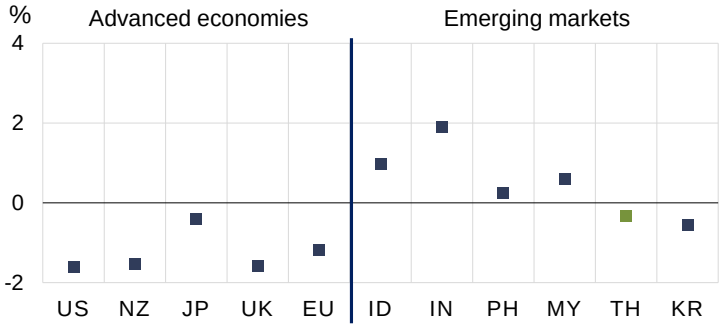
Note: * Business loans covers lending activities of Other Depository Corporation (ODCs), Other Financial Corporations (OFCs), and non-resident
** The definition of private credit was redefined to cover broader types of lenders and financial instruments. The data was first published in the MPR in 2019Q4, while the data was available from January 2012.

Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Since the beginning of the first quarter, the Bank of Thailand, the government, related agencies, and financial institutions had been introducing a series of measures to alleviate debt servicing burden and enhance liquidity for businesses and households affected by unfavorable economic conditions (**Box**: *Credit assistance measures to support borrowers* in Chapter 3). In addition, the MPC cut the policy rate twice in February and March to ease financial conditions further. The real policy rate declined from the previous quarter and remained at a low level relative to those of other emerging markets (EMs) (Chart 2.14).

Chart 2.14 Thailand's real policy rate decreased due to the policy rate cut and remained at a low level relative to those of other EM countries.

Real policy rates*



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to the survey by Consensus Economics on 9 March 2020 (data as of 24 March 2020)

Sources: Bloomberg and Consensus Economics, calculation by Bank of Thailand

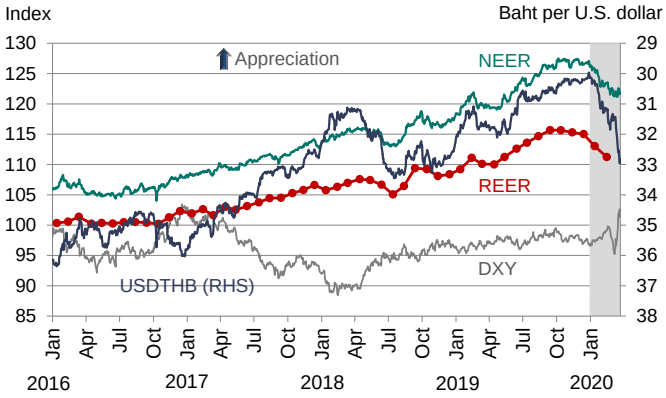
[9/] Results of financial institutions' survey as published in the Report on Credit Conditions Q4/2019 and Outlook for Q1/2020,

The baht against the U.S. dollar and the nominal effective exchange rate (NEER) index depreciated due to the COVID-19 outbreak.

The baht against the U.S. dollar in the first quarter of 2020 depreciated significantly relative to the end of the previous quarter (Chart 2.15). At the beginning of the quarter, the baht depreciation was in line with risk-off sentiments caused by concerns over the COVID-19 outbreak as the number of infections and deaths continued to rise in China. The baht depreciation was also driven by better-than-expected U.S. economic data outturns and market expectations of a policy rate cut ahead of the MPC meeting on February 5, 2020. Subsequently, the baht weakened rapidly toward the end of the quarter in line with regional currencies after the WHO declared COVID-19 as a pandemic. Foreign investors in need of U.S. dollar liquidity thus continued to sell off their risky assets, including regional and Thai assets. The MPC's policy rate cut at the special meeting on March 20, 2020 also prompted the baht to weaken somewhat. The baht averaged at 31.93 baht to U.S. dollar in March 2020, a 5.4 percent depreciation from an average of December 2019.

Chart 2.15 The Baht depreciated against the U.S. dollar as non-residents sold off EMs assets to hold low risks assets

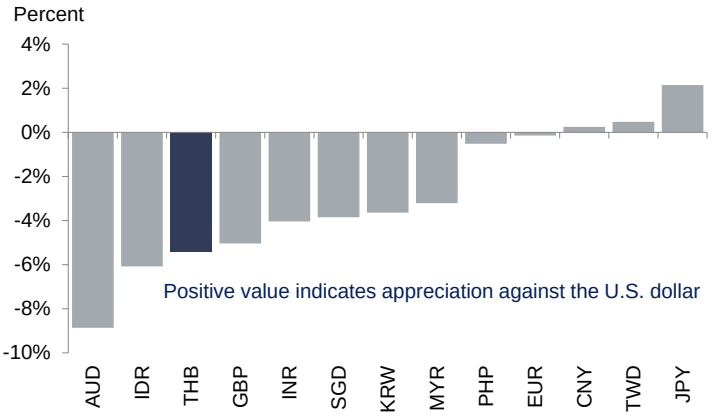
USDTHB, NEER, DXY



Source: Bank of Thailand, Bloomberg and Reuters (data as of 24 March 2020)

The nominal effective exchange rate (NEER) index averaged at 121.9 in March (as of March 24, 2020), a depreciation of 3.8 percent from December 2019. This was attributable to the baht depreciation against most trading partner currencies (Chart 2.16). Meanwhile, the preliminary **real effective exchange rate (REER)** in February weakened 3.3 percent from December 2019. Going forward, exchange rates would likely remain volatile due to high uncertainties surrounding the COVID-19 pandemic, monetary policy and additional economic stimulus measures of major advanced and regional economies, as well as investor confidence towards emerging market assets.

Chart 2.16 The Baht depreciated against the U.S. dollar in line with movements of most regional currencies
(Avg. in Mar 20 compared to avg. in Dec 19)



Source: Bank of Thailand and Reuters (data as of 24 March 2020)

The COVID-19 outbreak led to heightened risks to financial system in the short run and greater investors' concerns, resulting in rapid redemption of investment units in some mutual funds. Looking ahead, debt servicing ability of households and businesses would deteriorate, which could restrict decisions of financial institutions to extend loans. Should credit extension slow down considerably, this would in turn affect economic activities, resulting in a negative feedback loop.

Thailand's financial stability risks increased in the short term due to the COVID-19 outbreak. Nevertheless, Thailand's financial institutions maintained robust financial positions, as reflected by adequately high levels of commercial banks' capital buffers and reserves, providing cushion against risks of deteriorating credit quality. Likewise, external stability remained sound, as indicated by high level of international reserves, low external debt to GDP, and sustained current account surpluses. This would be adequate to cushion against the U.S. dollar liquidity squeeze in the global financial markets. In addition, Thailand's public debt remained low, providing room for borrowing to finance greater government spending in case of emergency. Nevertheless, there remained other risks that warranted monitoring:

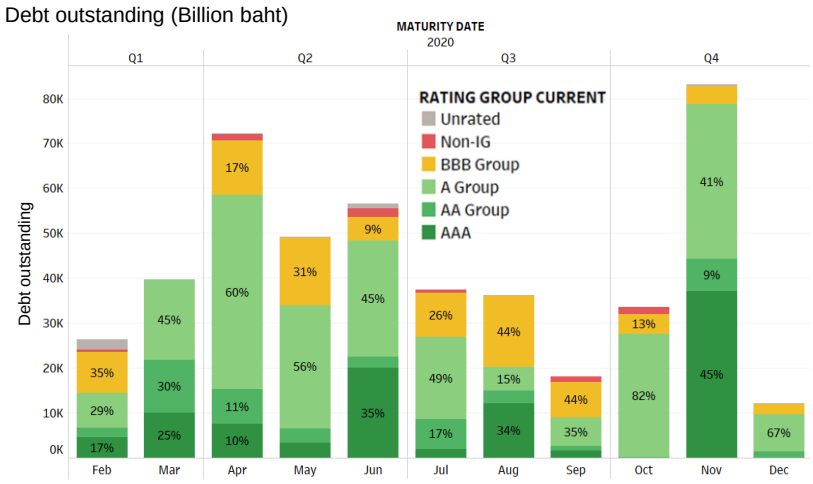
1. Risks in the bond markets increased as some investors rushed to redeem their investment units from some mutual funds. In March, foreign asset prices were excessively volatile due to the COVID-19 pandemic, affecting net asset values (NAVs) of some mutual funds in Thailand with majority of investment in foreign debt instruments. Some major investors therefore rushed to redeem their investment units from these funds, especially daily fixed income funds (Daily FI), which allowed for daily redemption and invested in foreign assets. As a result, these mutual funds became short of liquidity and had to sell off their government bonds prior to their maturities in order to have enough cash to return to unitholders. Consequently, prices of government bonds fell, in turn affecting NAVs of other mutual funds and encouraging investors to redeem their investment units from these mutual funds. If this problem was not urgently addressed in order to restore financial market stability, the problem could exacerbate and pose systemic risks to the functioning of the Thai financial markets. In particular, if mutual funds could not generate enough liquidity from selling off government bonds, they might be forced to sell off corporate bonds. This would impact both prices and liquidity of corporate bond markets, as well as NAVs of other mutual funds.

2. Corporate bond default risks increased, which could trigger knock-on effects on other related parties in the financial system. Falling corporate income due to impacts of the COVID-19 and a sharp fall in oil prices could heighten default risks or rollover risks of Thai companies that raised funds through the bond markets. This was as investors have lower risk tolerance. In terms of credit rating, most Thai corporate bonds maturing in 2020 were rated A and BBB. The latter group was vulnerable to being downgraded to non-investment grade should the issuing companies experience impacts from the economic contraction (Chart 2.17). Nevertheless, if corporate bond default risks continued to rise, it would trigger knock-on effects to other related parties in the financial system. These included cross-default, especially in the case that creditors were commercial banks, and some saving cooperatives with high corporate bond holdings to total assets.

In light of the above financial stability risks, the Bank of Thailand continued to support liquidity provision in the bond markets through government bond purchase program, the policy rate cut, and joint announcement of measures to stabilize the Thai

financial markets in collaboration with other related agencies. These were preventive measures against systemic risks, including measures to support mutual funds affected by liquidity shortage in the financial markets and the establishment of the Corporate Bond Stabilization Fund (BSF) to mitigate rollover risks in the corporate bond markets (**Box***: Measures to stabilize the Thai financial markets* in Chapter 3).

Chart 2.17 Most corporate bonds maturing in 2020 were rated A or BBB. The latter group was vulnerable to being downgraded to non-investment grade due to the impact of economic slowdown



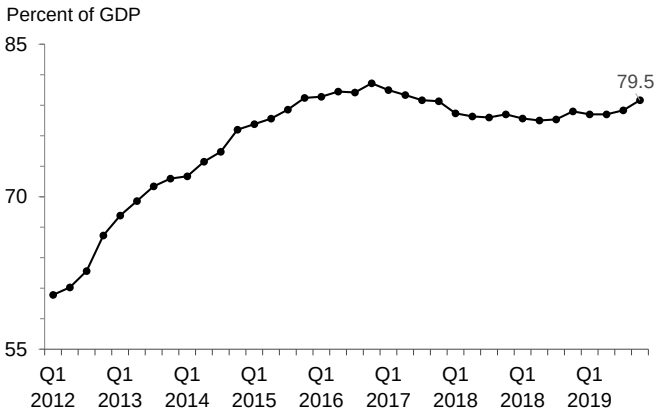
Source: ThaiBMA, compiled the Bank of Thailand

3. Weakening ability of households and businesses to manage liquidity and service debt amid the economic slowdown would affect credit extension and economic activities, resulting in a negative feedback loop.

Prior to the spread of COVID-19, household debt to GDP ratio in the fourth quarter of 2019 rose to 79.5 percent (Chart 2.18). Meanwhile, the quality of household credit was expected to deteriorate due to income shocks, such as the drought, which would raise default risk of the household sector. The NPL ratio of consumer loans in the fourth quarter of 2019 also increased from the previous quarter to 2.9 percent, mainly in line with deteriorating quality of mortgage loans and personal loans with housing as collateral.

Chart 2.18 Household debt* exhibited an increasing trend.

The ratio of household debt to GDP**



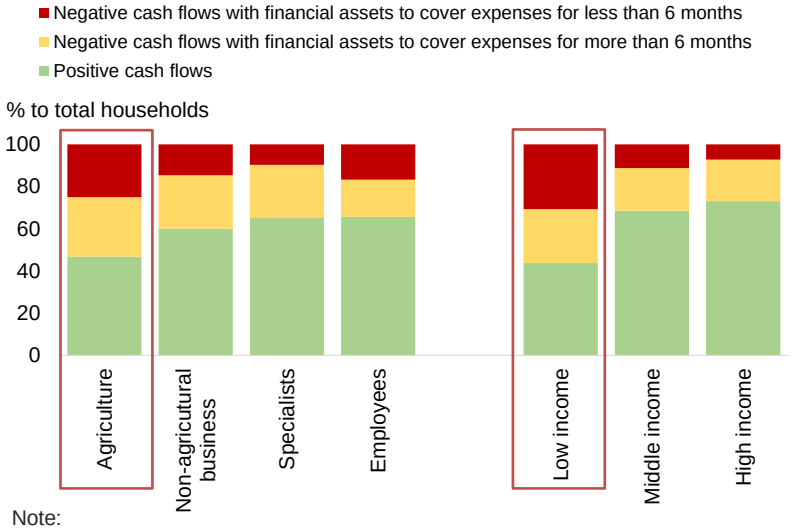
Note: *Loans to households by financial institutions
**Calculated by averaging the 4 latest quarterly GDP
Household debt and GDP data are revised, resulting in the difference in household debt to GDP ratios compared with the previous MPR.

Source: Bank of Thailand

An analysis on household financial positions, based on household socio-economic survey in the first three quarters of 2019, showed that agricultural households and households with monthly income of less than 30,000 baht were the most vulnerable with high default risks. This was because these households had higher proportion of negative cash flows relative to other groups. Some of these households had financial assets that could only cover their expenses for less than 6 months, resulting in limited ability to cope with income shocks (Chart 2.19). This situation also reflected the outlook of household debt in some categories that might increase in the period ahead, as some households would face liquidity problems and thus need to borrow to cover their spending.

Chart 2.19 Farmers and low-income employees had low capacity to absorb income shocks

Proportion of households classified by financial positions



Note:
(1) Cash flow is calculated from monthly income minuses total expenses and monthly debt burden, where negative cash flows reflects insufficient income for expenses each month.
(2) The number of months that households survive or have enough financial assets to liquidate to support negative cash flows is calculated households' financial assets divided by negative cash flows. The calculation is under the assumption that households need to liquidate assets in order to cover the negative cash flows.

Source: Socio-Economic Survey (SES) Q1-Q3 2019, calculated by Bank of Thailand.

Overall financial position of businesses became more vulnerable, especially SMEs. This was reflected by the sustained decline of operating profit margin (OPM) and negative interest coverage ratio (ICR) of some enterprises.[10/] Meanwhile, credit quality of business loans deteriorated across almost all categories, having extended to include larger enterprises. The NPL ratio of SME loans remained high at 4.6 percent in the fourth quarter of 2019.[11/]

Looking ahead, the spread of COVID-19 could cause a severe impact on the revenue of some businesses, especially those related to tourism, transportation and some segments of the manufacturing sector. This would have repercussions for employment, debt servicing ability, and credit extension of financial institutions. This would in turn affect economic activities, resulting in a negative feedback loop. However, the Bank of Thailand and the government, in

[10/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies, which remained in the negative territory.
[11/] The NPL ratio of SME loans from 2013 to date averaged at 4.0 percent.

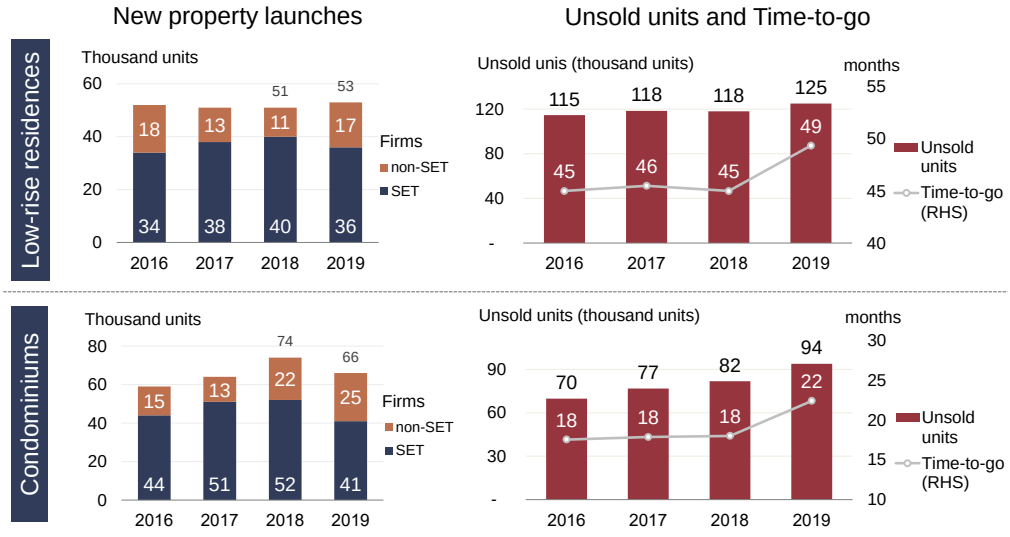
collaboration with financial institutions and financial service providers, introduced measures to assist borrowers affected by the economic conditions. This was expected to help relieve the impact to a certain level **(Box:** *Credit assistance measures to support borrowers* in Chapter 3).

4. Oversupply in the real estate market would increase due to weaker demand for residences from domestic and foreign buyers.

In January 2020, the Bank of Thailand relaxed the loan-to-value (LTV) measure in order to facilitate home ownership. The relaxed conditions included, first, for the first mortgage contract of residential units with value under 10 million baht, homebuyers would be allowed to borrow an extra 10 percent above the collateral value to cover necessary moving expenses. Second, for the second mortgage contract, homebuyers would be subject to 90-percent LTV ratio provided that the first contract had been serviced for at least two years, and 80-percent LTV ratio if the first contract had been serviced for less than two years (from three years in the previous regulation).[12/]

Going forward, the real estate market would experience greater risks due to substantial oversupply. At the end of 2019, unsold units of low-rise residences and condominiums increased from 2018 by 5.9 and 14.6 percent respectively. This corresponded to the longer time-to-go from 45 to 49 months for low-rise residences, and from 18 to 22 months for condominiums. This occurred as developers continued to launch new projects despite weakening demand, particularly small- and medium-sized developers that were not listed in the stock exchange and able to adjust themselves slower than large, listed developers. (Chart 2.20).

Chart 2.20 Risk in real estate market increased from oversupply and a longer time-to-go in both low-rise residences and condominiums

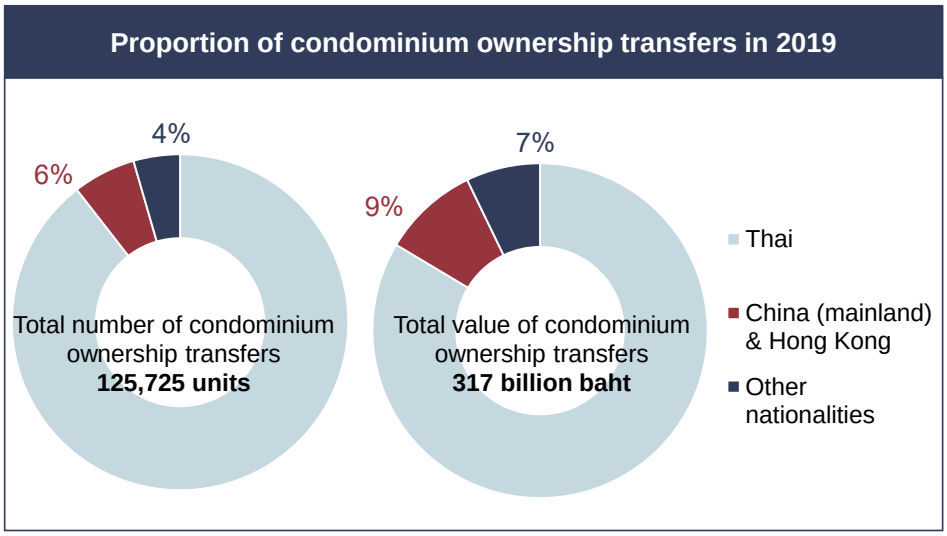


Source: AREA, calculated by Bank of Thailand

[12/] For the first mortgage contract for residential units with value exceeding 10 million baht, the LTV ratio was adjusted to 90 percent (from 80 percent previously).

In addition, the real estate market was impacted by the COVID-19 outbreak as residential demand from domestic and foreign buyers continued to weaken from the previous year. However, the impact of falling foreign demand was expected to be limited because foreign buyers accounted for only 10 percent of total ownership transfers, or 16 percent of the total value of condominium transfers in 2019 (Chart 2.21).

Chart 2.21 Overall impact from a fall in foreign demand due to the COVID-19 could be limited as foreign buyers' proportion to total condominium ownership transfers remained small



Source: Real Estate Information Center (REIC), calculated by Bank of Thailand

2.2 Outlook for the Thai Economy

The Thai economy would contract significantly, expanding at a much lower rate than the assessment in the previous *Monetary Policy Report*. Headline inflation would be negative in line with energy prices.

The Thai economy would contract significantly, expanding at a much lower rate than the assessment in the previous *Monetary Policy Report.* The Thai economy was expected to contract 5.3 percent in 2020 but would recover and expand 3.0 percent in 2021, despite at a level below its potential throughout the forecast horizon. The spread of COVID-19, which was declared as a pandemic, would have a severe impact on Thailand's trading partner economies, causing some economies to enter into recession. The spread of COVID-19 also caused supply chain disruptions in several countries, affecting the Thai economy through multiple channels. In particular, tourism and merchandise exports were severely affected. As a result, some businesses and households that relied on foreign sources of income received lower revenue and faced liquidity problems. Business and consumer confidence also fell to the lowest levels in many years, triggering knock-on effects on domestic economic activities, both consumption and investment. Moreover, farm income would still be depressed by the drought, while public expenditure would experience a larger impact than previously assessed due to the delayed enactment of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020).

Nevertheless, **the government's implemented measures[13/] would be the key mechanism in alleviating the impact on the economy,** by providing support to vulnerable businesses and households to some extent. However, the measures would be unable to offset a significant contraction in aggregate demand. The Thai economy would therefore contract substantially in 2020, which was the first contraction since 2009. Moreover, the Committee assessed that the tourism sector would take some time to begin to recover, and would not return to its previous growth path in the following periods.

Additional assumptions underlying the economic projection in this *Report* include (1) the COVID-19 outbreak in Thailand would be contained within the second quarter of 2020 according to the Ministry of Public Health estimate, and (2) the projection took into account measures implemented before March 25, 2020, not including any measures thereafter.

Headline inflation would be negative due to the sharp decline in energy prices in line with crude oil price assumptions. Headline inflation was projected to be negative at -1.0 percent in 2020, before gradually increasing to 0.3 percent in 2021 consistent with the gradual economic recovery outlook. However, headline inflation would remain below the inflation target.

Table 2.2 Forecast summary

Percent	2019*	2020	2021
GDP growth	2.4	-5.3 (2.8)	3.0
Headline inflation	0.7	-1.0 (0.8)	0.3
Core inflation	0.5	-0.1 (0.7)	0.1

Note: * Outturn
( ) Monetary Policy Report December 2019
Sources: NESDC, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to slow down significantly in 2020 as several economies would enter into recession due to the COVID-19 outbreak. However, trading partner economies would expand in 2021, partly owing to the very-low-base effect, benefits from financial and fiscal measures in many countries, and improving COVID-19 situation.
- The federal funds rate was cut twice in March 2020 to 0.00-0.25 percent to support the economy and the financial system that were severely affected by the COVID-19 outbreak. The federal funds rate was projected to remain at this level throughout the forecast horizon.
- Regional currencies (excluding the Chinese yuan) depreciated throughout the forecast horizon. This was because regional economies were affected by the COVID-19 outbreak before other regions, which led to deterioration of investor confidence in holding regional securities. This resulted in capital outflows from regional stock markets, although the situation would start to improve in 2021.

[13/] Government measures that were announced before March 25, 2020.

- Dubai crude oil prices were revised down substantially throughout the forecast horizon due to both the decline in global demand as a result of the COVID-19 outbreak and the decision of Saudi Arabia, which was the largest oil producer, to lower crude oil prices. This was following the failure of the OPEC and Russia to agree on the crude oil production cut, causing crude oil prices to sink to the lowest level in 17 years.
- Farm income, excluding government support measure, was revised down in 2020 in line with the decline in agricultural output, especially off-season rice and sugarcane, mainly caused by a larger-than-expected impact of the drought in the first half of the year. Meanwhile, agriculture prices were expected to rise, especially prices of palm oil, glutinous rice and sugarcane. In addition, farm income in 2021 was expected to increase due to the lower base of agricultural output in 2020 and better assumptions on the weather and dam water levels than in 2020.
- Public spending at current prices, both public consumption and investment, was lower than previously assessed on account of a larger-than-expected impact of the delayed enactment of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020), the change in the structure of the annual budget expenditure for fiscal year 2021, as well as a smaller investment budget and a prolonged delay of some state-owned enterprise (SOE) investment.

Table: Summary of forecast assumptions

Annual percentage change	2019*	2020	2021
Dubai crude oil price (U.S. dollar per barrel)	63.5	35.0 (63.0)	34.0
Farm income (% YoY)	3.3	-2.5 (-0.8)	5.0
Government consumption at current price (billion baht)[1/]	2,723	2,863 (2,889)	2,993
Public investment at current price (billion baht)[1/]	966	1,040 (1,067)	1,141
Fed funds rate (% at year end)	2.125	0.125 (1.625)	0.125
Trading partners' GDP growth (% YoY)[2/]	2.7	0.2 (2.7)	2.0
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	156.6	158.7 (157.4)	158.7

Notes: [1/] Assumption includes spending on infrastructure investment plans

[2/] Weighted by each trading partner's share in Thailand's total exports

[3/] Increasing index represents depreciation, decreasing index represents appreciation

* Outturns

( ) Monetary Policy Report December 2019

Merchandise exports would decline markedly in the short term due to the significant global economic slowdown.

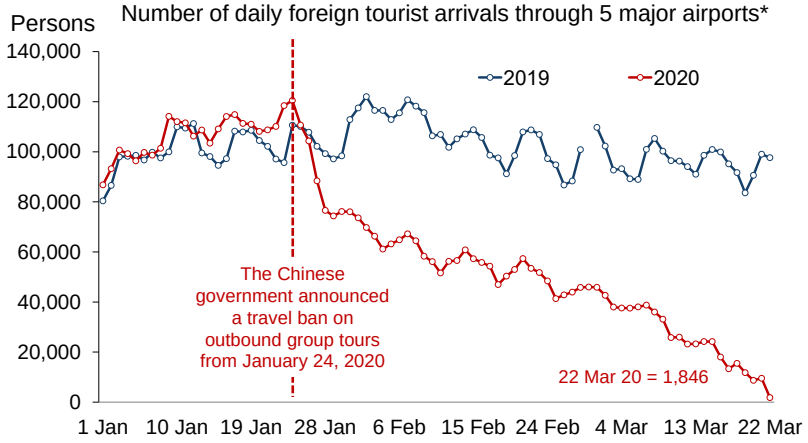
The value of merchandise exports was expected to markedly shrink both in terms of volume and prices. Global trade volume would contract in response to global economic conditions, as several countries would enter into recession as a result of the COVID-19 outbreak. Moreover, Thai merchandise exports would be additionally affected by supply chain disruptions, as well as lower demand from oil-exporting countries on account of the substantial fall in crude oil prices. Nonetheless, exports of some products would benefit from rising external demand, despite being unable to offset overall contraction in merchandise exports. In particular, the benefit would be seen in necessities such as rice and canned fish, and medical equipment such as rubber gloves and cassava, which was a raw material for alcohol production. Meanwhile, prices of merchandise exports would significantly decline in

tandem with crude oil prices. Prices of export products, which were sensitive to crude oil prices, such as petroleum products, and petrochemical and chemical products, would be significantly lower than previously assessed. Nonetheless, the value of merchandise exports was projected to expand slightly in 2021, on account of the gradual recovery of the global economy and global trade volume once the spread of COVID-19 subsided. **The Committee assessed that the value of merchandise exports would contract 8.8 percent in 2020, before expanding slightly at 0.2 percent in 2021.**

Exports of services were expected to severely contract in the short run due to the COVID-19 outbreak

Exports of services would severely contract as the COVID-19 outbreak led to travel restrictions, including flight suspension and border closure in several countries. Many airlines then had to temporarily suspend flight services, resulting in decline in foreign tourist figures of all nationalities (Chart 2.22). In addition, the significant economic slowdown in several countries due to the COVID-19 outbreak led to declines, not only in foreign tourist figures, but also in tourist spending per head. However, foreign tourist figures were expected to gradually increase in the second half of 2020, based on the assumption that the spread of COVID-19 would be contained within the second quarter of 2020. Nevertheless, tourists would start to travel once the vaccines for the coronavirus were to be widely distributed. **The foreign tourist figures were projected to be 15.0 and 20.0 million persons in 2020 and 2021, respectively.**

Chart 2.22 Foreign tourist figures were expected to fall significantly



Note: *5 major airports include Suvarnabhumi, Don Mueang, Chiang Mai, Hat Yai and Phuket International Airports, which account 83 percent of total foreign tourists.

Source: TAT Intelligence, Ministry of Tourism and Sports, compiled by Bank of Thailand.

The current account balance was projected to record a surplus of 19.4 and 19.2 billion U.S. dollars in 2020 and 2021, respectively. The services balance was revised from being in surplus to deficit, as tourist receipts declined considerably in line with the decline in the number of tourists and spending per head. Meanwhile, the trade balance was expected to register a larger surplus due to a substantial contraction in imports value. Import volume contracted in line with imports of raw materials and intermediate goods for export-related manufacturing, and capital goods for domestic investment. Meanwhile, import prices declined in tandem with global crude oil prices.

Private consumption would contract in the short term despite government relief measures, but would expand in the subsequent period in line with the gradual recovery in tourism and merchandise exports.

Private consumption was expected to contract in 2020, which would be the first contraction since 2009. The COVID-19 outbreak led to severe contraction of tourism sector. In addition, suspension of a wide range of economic activities and the enactment of the emergency decree from March 26, 2020, affected consumption in the services and away-from-home spending categories. Meanwhile, non-farm income decreased substantially because the manufacturing sectors were affected by the sharp decline in both domestic and external demand. Moreover, farm income was expected to decline due to a more-severe-than-expected impact of the drought, reducing capacity of the agricultural sector to absorb unemployed workers moving from manufacturing and services sectors. However, government measures to alleviate the impact of the COVID-19 outbreak would partly help relieve the impact on private consumption to a certain level. These included measures on credit, transfers, expenses reduction, and taxes.[14/] In the period ahead, private consumption was projected to gradually edge up in line with the recovery of tourism and merchandise exports once the spread of COVID-19 could be contained. However, the slow recovery of household income and the elevated household debt would continue to weigh on private consumption in the period ahead.

Private investment was projected to contract in the short term in line with both domestic and external demand.

Private investment was projected to contract in 2020 in line with both domestic and external demand due to the COVID-19 outbreak. This led profitability of businesses and investment confidence to deteriorate markedly. However, private investment was expected to improve in 2021 thanks to a gradual recovery in merchandise exports, the PPP projects in infrastructure investment projects that had been postponed from previous periods, as well as a greater clarity pertaining to investment in the 5G network following the auction in the first quarter of 2020.

Public expenditure would be a key driver of economic growth

Public expenditure, both consumption and investment, was expected to expand and would be a key driver of the Thai economy. Following the enactment of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020), although government agencies adjusted the procurement process for a faster disbursement, this could not offset a larger-than-expected impact of the delayed enactment of the Annual Budget Expenditure Act in the previous period. SOE investment was revised down in line with project postponement and a lower budget for capital expenditure on transportation projects. This included, for example, the Bangkok-Nakhon Ratchasima high-speed rail project, whose environmental impact assessment (EIA) report was being revised. Meanwhile, **the first and second packages of government measures to mitigate the impact of the COVID-19 outbreak, totaling 464 billion baht or 2.8 percent of GDP (excluding tax relief measures worth approximately 260 billion baht or 1.5 percent of GDP)**, **were mostly measures on credit and transfers. This was expected**

[14/] Government measures announced prior to 25 March 2020

to directly cushion private consumption and investment, not through public consumption and investment (**Box**: *Government relief measures to alleviate the impact of COVID-19* in Chapter 2). **In sum, public investment would expand at a lower rate than previously assessed in the previous *Monetary Policy Report,* while public consumption growth would remain largely unchanged.** Going forward, factors that would affect public expenditure included changes in the structure of the annual budget expenditure for the fiscal year 2021, resulting in changes in budget allocations for current and capital expenditures, and a lower budget expenditure in the period ahead as revenue collection was expected to be lower than previously assessed.

Projection of headline inflation would be negative in the short term in line with declines in energy prices and core inflation.

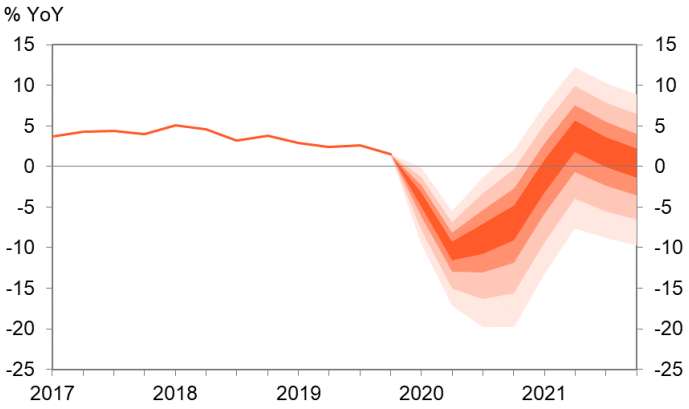
Projection of headline inflation in 2020 would be negative at -1.0 percent, which would be lower than the inflation target and significantly below the forecast in the previous *Monetary Policy Report.* The downward revision was consistent with a sharp decline in crude oil prices in early 2020, which led to a large fall in energy prices. Meanwhile, core inflation was expected to be negative on the back of a large decrease in demand-pull inflationary pressures. In addition, measures to reduce living expenses were expected to partly prevent prices from rising. **Nonetheless, headline inflation was projected to gradually increase to 0.3 percent in 2021 in line with the assumption of a gradual rise of crude oil prices. Core inflation was expected to edge up slightly from -0.1 percent in 2020 to 0.1 percent in 2021,** consistent with the economic recovery outlook. However, headline inflation would remain below the inflation target throughout the forecast horizon.

Economic and inflation projections were subject to greater downside risks and were highly uncertain.

The Committee assessed that economic and inflation projections were highly uncertain and mainly depended on the situation of the COVID-19 spread and the success of the containment measures. Downside risks to the Thai economy increased significantly from the assessment in the previous *Monetary Policy Report* (Chart 2.23). **The possibilities that the Thai economy would underperform the baseline projection would be due to the following factors.** First, the global economy could enter into a deep recession should the outbreak become more severe and prolonged than expected, exerting additional adverse impacts on Thailand's merchandise exports. Second, global financial stability risks might affect the Thai economy in the medium term. Third, Thailand's tourism might not recover in the second half of 2020 if the outbreak in Thailand and other countries prolonged and could not be contained by the second quarter of 2020. Fourth, risk of a wider spread of COVID-19 in Thailand might force the government to implement stricter measures to control the outbreak. As a result, this might cause some businesses to permanently shut down operations or significantly reduce employment, thereby impacting household income. Fifth, more-severe-than-expected impact of the drought could affect farm income and reduce the capacity of agricultural sector to absorb workers moving from other sectors. However, **there were possibilities that the Thai economy would outperform the baseline projection due to** (1) a faster-than-expected development of vaccines and medicines for the COVID-19, (2) better-than-expected impacts of additional government's stimulus measures to alleviate the impact and support domestic consumption and investment, as well as measures to assist

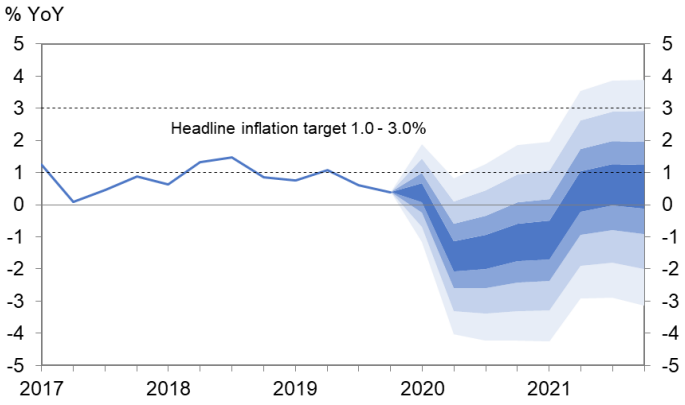
borrowers, and (3) a higher-than-expected increase in domestic demand as a result of public infrastructure investment and PPP projects, which might incentivize the private sector to invest more and faster. Moreover, **there remained downside risks to headline and core inflation projections** in line with risks to crude oil prices and the economic growth projection (Charts 2.24 and 2.25).

Chart 2.23 Growth forecast



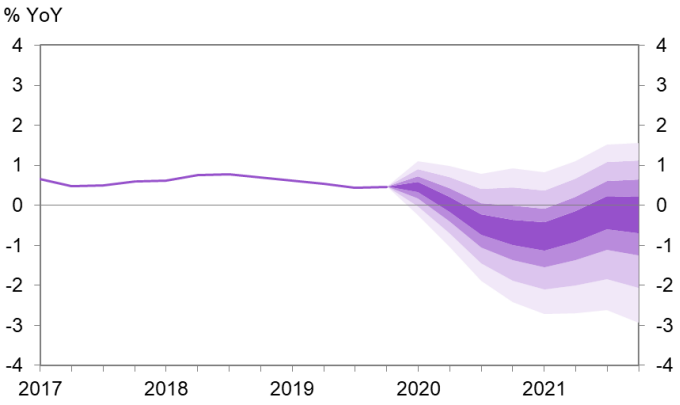
Note: Fan chart covers 90% of the probability distribution

Chart 2.24 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.25 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2019*	2020	2021
GDP growth	2.4	-5.3 (2.8)	3.0
Domestic demand	3.3	-0.8 (3.2)	2.6
Private consumption	4.5	-1.5 (3.0)	2.1
Private investment	2.8	-4.3 (3.4)	2.2
Government consumption	1.4	2.6 (2.6)	2.2
Public investment	0.2	5.8 (6.3)	8.1
Exports of goods and services	-2.6	-16.4 (1.4)	5.0
imports of goods and services	-4.4	-11.6 (1.6)	4.3
Current account (billion, U.S. dollars)	37.9	19.4 (30.0)	19.2
Value of merchandise exports	-3.2	-8.8 (0.5)	0.2
Value of merchandise imports	-5.4	-15.0 (1.4)	-0.4
Number of foreign tourists (million person)	39.8	15.0 (41.7)	20.0

Note: *Outturns

( ) Monetary Policy Report December 2019

Government relief measures to alleviate the impact of COVID-19

Summary of government relief measures to alleviate the impact of COVID – 19

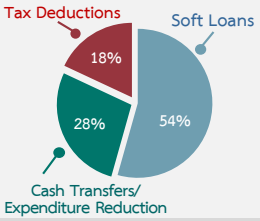
Phase I and II measures following the cabinet resolutions on March 10 and 24, 2020

Overview

Government injected money into the economy through Phase I and II measures, totaling

464.1 billion (bn.) baht (2.8% of GDP)

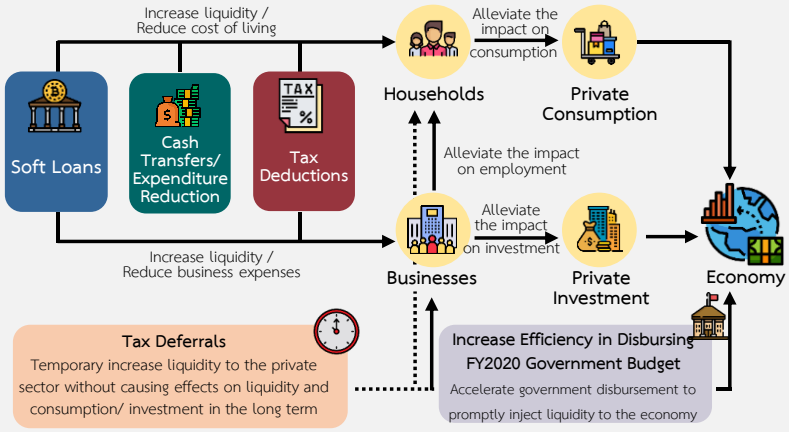
(excluding tax deferral measures)



Target Groups

Households | Businesses | Households & Businesses

Transmission of liquidity to the economy



Measures	The First Package	Budget (bn.)	Period	The Second Package	Budget (bn.)	Period
Soft Loans 252 bn. baht (1.5% of GDP)	**Soft loans** for 2 years at 2% annual interest rate with max credit limit of 20 million baht/business	150	Mar – Dec 30, 2020	**Emergency loans for self-employed workers** for 2½ years at 0.1% monthly interest rate with max limit of 10,000 baht/person	40	Mar 24 - Dec 30, 2020
	Loans to support employment for 3 years at 3% annual interest rate for businesses registered with the Social Security Office (SSO) (business must maintain at least 80% of its employment from the time the loans are issued up to 3 years.)	30		**Special credit facilities for employees** for 3 years at 0.35% monthly interest rate with max limit of 50,000 baht/person	20	Mar 24 - Dec 30, 2020
				Soft Loans for low-income group at 0.125% monthly interest rate for 2 years	2	
				Soft Loans for SMEs for 5 years with a limit of 3 million baht/SME, with a concessional rate of 3% per annum for the first 2 years	10	Mar 24 - Dec 30, 2020
Cash Transfers/ Expenditure Reduction 128 bn. baht (0.8% of GDP)	**Refund of electricity and water provider deposits**	36	Mar 2020 (Electricity) May 2020 (Water)	**Compensation for workers (outside of the social security system)** 5,000 baht/person/month for 3 months	45[3/]	Apr - Jun 2020
	Reduction of required contributions to the Social Security Fund (SSF) for employers and employees[1/] to 4%/month (section 33) and to 7%/month (section 39)	17	Mar - Aug 2020	**Training for employees affected by the COVID-19 outbreak** support the training and expenses to participants	4	
	Reduction of electricity and water bills	6	Apr - Jun 2020			
	COVID-19 relief funds	20[2/]				
	Reduction of fees, rental fees, charges levied by government agencies and state-owned enterprises					
Tax measures						
1. Tax Deductions 84.1 bn. baht (0.5% of GDP)	**Additional 200,000-baht income tax deduction for investment in Super Saving Fund (SSF)** from the original cap and not subjected to deduction limit of retirement saving fund (for investment in at least 65% of its net asset value in shares listed on the SET)	14	Apr - Jun 2020	**Increase in tax deduction limit for health insurance premiums to 25,000 baht** (The total deduction, combining deduction on life insurance premiums and living allowance deposits, must not exceed 100,000 baht.)	3	Tax year 2020 onwards
				Income tax exemption on hazard pay for medical personnel appointed by the Ministry of Public Health	0.1	Tax year 2020
	Additional income tax deduction for SMEs - 3 times the wage expenses for SMEs	64	Apr - Jul 2020	**Exemption of import duties on products related to prevention and treatment of the COVID-19**	1	Mar 24 - Sep 30, 2020
	- 1.5 times the interest expenses for SMEs enrolling in the Phase I of soft loan program	1	Apr - Dec 2020	**Promoting debt restructuring** Tax exemption and reduction on transfer and mortgage fees for condominium and properties related to debt restructuring	1	Jan 2020 - Dec 2021
2. Tax Deferrals Temporary impact on government cash flows, but not on revenue collection for the whole year	**Acceleration of VAT refund process**		Apr - Sep 2020	**Extension of personal income tax filing deadline** : from Jun 30, 2020 → Aug 31, 20 20		Apr – Jun 2020
	Reduction of withholding tax : to 1.5% for any payment method : to 2% for payment via e-withholding tax (The reduction amount will be included in the half-year and annual income tax payments.)		Oct 2020 - Dec 2021	**Extension of corporate income tax filing deadline** - P.N.D 50 (TY 2019) : Apr 1 – Aug 30, 2020 → Aug 31, 2020 - P.N.D 51 (TY 2020) : Jul 1 – Sep 29, 2020 → Sep 30, 2020		
				Extension of excise tax filing deadline for affected operators : from Mar 1 – May 31, 2020 → Jul 15, 2020		
				Extension of filing deadline for taxes under the purview of the Revenue department for businesses affected by the COVID-19		
				Extension of excise tax filing deadline for oil and petroleum products : from within 10 days → within the 15th of the following month		
Government Expenditures	• **Acceleration of budget disbursement on any procurement** under 2 million baht • **Adjustment in the procurement process** for a faster disbursement • **COVID-19 relief funds**: procurement and distribution of masks and disinfectants[2/]			The measures will accelerate disbursement of the FY2020 Government Budget, not from additional budget		

Note : 1/ Excluding further announcement on Mar 24 and 31, 2020. According to the notification of the Ministry of Labor and Social Welfare, the Thai government announced additional measures of reducing contributions to the social security fund for employees to 1%/month (section 33) and to 86 baht/month (section 39) for 3 months (Mar - May 2020)
2/ These two measures will be covered by the same budget, but details of budget allocation have not yet made official.
3/ Excluding results of the cabinet resolution on Mar 31, 2020, where the target groups were expanded to 9 million people with the budget increased to 135 bn. baht.

Source : (1) The Thai Government's Phase I and II measures to alleviate the impact of the COVID-19 on the Thai economy approved by the Cabinet on Mar 10 and 24, 2020
(2) The cabinet resolution on Mar 17, 2020 (3) Notification of the Ministry of Finance No. 22/2020 and No. 26/2020 (4) Related news

3. Monetary Policy Decision

The MPC cut the policy rate 2 times to 0.75 percent in order to alleviate the negative impacts, particularly from the COVID-19 pandemic which resulted in severe impacts on the economy and financial stability.

At the meeting on February 5, 2020, the Committee voted unanimously to cut the policy rate by 0.25 percentage point from 1.25 to 1.00 percent. The Committee assessed that the Thai economy would expand at a much lower rate than the previous forecast and much further below its potential due to the COVID-19 outbreak, the delayed Annual Budget Expenditure Act, and the drought. Headline inflation was projected to be below the lower bound of the inflation target throughout the forecast horizon. Financial stability became more vulnerable due to the prospect of an economic slowdown. In this situation, there was an urgent need to coordinate financial and fiscal measures. The Committee viewed that a more accommodative monetary policy stance would alleviate the negative impacts. Monetary accommodation would also support liquidity provision and debt restructuring for businesses and households severely affected from the economic slowdown, both of which should be urgently implemented and materialized.

Later on, the COVID-19 outbreak became more severe and widespread rapidly and continuously. This raised concerns among global financial markets, affecting the liquidity of the Thai financial markets during the week prior to the scheduled meeting at the end of March. **The Committee thus held the special meeting on March 20, 2020 and viewed that amid the rapid changes in the outbreak situation, prompt policy action prior to the scheduled meeting would help alleviate the impacts on the Thai economy and financial markets in a timely manner. The Committee thus voted unanimously to cut the policy rate by 0.25 percentage point from 1.00 to 0.75 percent,** effective on March 23, 2020. Nonetheless, the Committee viewed that the COVID-19 outbreak in the period ahead would be more severe than previously expected. Also, the situation would take some time before returning to normal, which would severely affect the Thai economy. Therefore, the policy rate cut would reduce interest burdens of affected borrowers, alleviate liquidity strain in the financial markets, mitigate the impacts on the economy, and would also reinforce fiscal measures already implemented and forthcoming.

At the meeting on March 25, 2020, the Committee voted 4 to 2 to maintain the policy rate at 0.75 percent. Two members voted to cut the policy rate by 0.25 percentage point, while one MPC member did not vote.[15/] The Committee viewed that the economic and inflation outlook remained consistent with the assessment at the special meeting. The Thai economy would markedly contract in 2020 due mainly to the COVID-19 outbreak. Headline inflation in 2020 would be negative. Financial system would be more vulnerable driven by contracting economy. Nevertheless, the Committee viewed that the policy rate cut at the previous special meeting, as well as the implemented measures to promote stability in the Thai financial markets, helped ease financial conditions. Commercial banks' lending rates gradually decreased and stability of domestic financial markets, especially the bond market, improved.

[15/] The MPC member attended the meeting through video conference to consider the economic outlook, financial stability, and the appropriate monetary policy formulation but did not vote.

Most members would wait to assess the adjustments in the financial markets as well as additional reduction in interest rates by financial institutions. Most members deemed it appropriate to preserve policy space to cushion against high uncertainties in the future. Meanwhile, all members viewed that the government's announced relief measures and loan repayment measures to assist borrowers affected by the COVID-19 were more targeted in addressing the problems. However, two members voted to cut the policy rate by 0.25 percentage point as the extra accommodative monetary policy would, in their views, be commensurate with the significant economic contraction.

In determining the appropriate course of monetary policy, details on the key assessment are as follows.

The Thai economy faced negative risk factors which remained highly uncertain and could severely affect the economy and financial stability.

The Committee viewed that **The Thai economy faced several negative risk factors that occurred in a similar period,** including the COVID-19 outbreak, the delay in the government's annual budget spending, and the drought. Such negative risk factors were highly uncertain and could result in a more severe and much broader impact than expected. The Committee thus assessed the situation through **scenario analysis**. In the worse scenario that negative risk factors coincided, there were possibilities that income of businesses and households would severely decrease, and debt servicing ability would deteriorate. Financial institutions would also experience higher NPLs and lower capital funds, causing financial institutions to be more cautious in extending loans. This led to difficulty in accessing funds and lower liquidity of the private sector, resulting in a **negative feedback loop** in the economic system.

In addition, **some pockets of risks to financial stability could become contagious if not urgently addressed.** Such risks included, for example, a reduction of business revenue that could increase default risks of debt-financing corporates or rollover risks. This could potentially trigger a cross default and if investor confidence declined, rollover risks could spill over to other corporate bonds. The Committee thus deemed it necessary to monitor risks and vulnerabilities closely and urged the BOT to prepare policy tools to be readily employed to maintain financial system stability given various situations. The Committee also supported the close coordination between the BOT and other related regulatory agencies to stand ready to prevent systemic risks in a pre-emptive manner.

In the short term, financial and fiscal measures should be coordinated to reduce risks to the economy and financial stability, addressing the problems proactively and targeted to prevent the contagion.

Thai government agencies coordinated both financial and fiscal measures to relieve the impacts and reduce the abovementioned risks, addressing problems in the following directions.

1. Policy conduct must be forward looking and pre-emptive, without having to wait for adverse impacts to be observed. For instance, during the first phase when most of the COVID-19 patients were contained in China and not widespread to other countries around the world, the Committee assessed that the spread of the COVID-19 had already affected the Thai economy, despite being highly uncertain whether there would be additional and a more severe impact. The Committee thus voted to cut the policy rate on February 5, 2020. Later on,

as the spread situation would likely be more severe with liquidity shortages in the Thai bond market, the Committee then held the special meeting on March 20, 2020 and cut the policy rate once again. In addition, the Committee encouraged the BOT to relax regulations so that financial institutions could proactively assist borrowers, including liquidity provision and debt restructuring, *(***Box***: Credit assistance measures to support borrowers* in Chapter 3*)* in order to prevent the impact of negative feedback loop on the Thai economy.

2. Measures must be targeted aiming to alleviate the impacts to businesses and the general public by coordinating both financial and fiscal measures. For instance, the BOT coordinated with financial institutions and other financial service providers to implement credit measures to assist borrowers affected by the COVID-19, issuing minimum measures to temporarily pause loan payment. Meanwhile, the government launched credit measures, cash transfer and expense reduction measures, as well as tax relief measures. *(***Box***: Government relief measures to alleviate the impact of COVID-19* in Chapter 2*)*. These measures were targeted to those experienced a direct impact.

3. Measures should aim at preventing the contagion, while ensuring the well-functioning of the financial system. For instance, the BOT ensured sufficient liquidity in the bond market through continuous government bond purchases, which helped reduce volatilities of government bond yields and prevent knock-on effects on other interest rates in the money markets. Nevertheless, on March 22, 2020, BOT and related agencies issued measures to support mutual funds affected by liquidity shortage in the financial markets and established the Corporate Bond Liquidity Stabilization Fund (BSF) to build confidence and reduce rollover risks in the bond market. (**Box:** *Measures to stabilize the Thai financial markets* in Chapter 3). Moreover, the Committee supported the increasing role of mechanisms to help reduce risks of NPLs. These mechanisms included, such as, the Thai Credit Guarantee Corporation that provided credit guarantee, and the Asset Management Company (AMC) that helped purchase non-performing assets out of financial institutions' balance sheet for future management.

The key challenge for the Thai economy and financial system was the distribution of liquidity in the financial system, which was ample at the macro level, to businesses and households.

The Committee viewed that the current challenges for the Thai economy and financial system were different from those during the financial crisis in 1997. At present, Thailand had a high level of international reserves, while financial institutions were sound with a high level of capital funds. Liquidity in the financial system remained adequate at the macro level. The key challenge was to urgently provide liquidity and credit supports to affected businesses and households. In particular, measures that allowed for debt moratorium, reduced loan installments, and extended debt repayment would help alleviate the immediate impacts. For the next step, debt restructuring must be urgently and constantly implemented in parallel with distribution of liquidity, which was ample at the macro level, to businesses and households in need. Examples of such measures could include provision of soft loans for businesses and households with the potential to repay their loans together with credit guarantees, so that loans could be allocated to SMEs more efficiently. Such measures could help those affected to get through this period.

In the long run, some pockets of risks in the financial system amid the lower-for-longer interest rate environment, as well as the stability of foreign exchange markets, must still be monitored.

The Committee assessed that concerns over the COVID-19 led investors to reduce their holdings of risky assets in the short run, as reflected by subdued activities in the real estate and stock markets. However, in the long run, as investors' concerns subsided, the search-for-yield behavior that could lead to underpricing of risks must be monitored. This behavior could become more severe in the lower-for-longer interest rate environment. In addition, although the baht depreciated to some extent, the Committee viewed that in the period ahead as the spread of the COVID-19 would subside, the baht would still face upward pressures from current account surplus which was expected to remain persistent. The Committee, therefore, supported the BOT to continue to relax regulations to promote capital flow balance. On February 28, 2020, the Ministry of Finance and the BOT agreed to increase the threshold for export proceeds that did not need to be repatriated from below 200,000 to 1 million U.S. dollars per bill of lading.[16/]

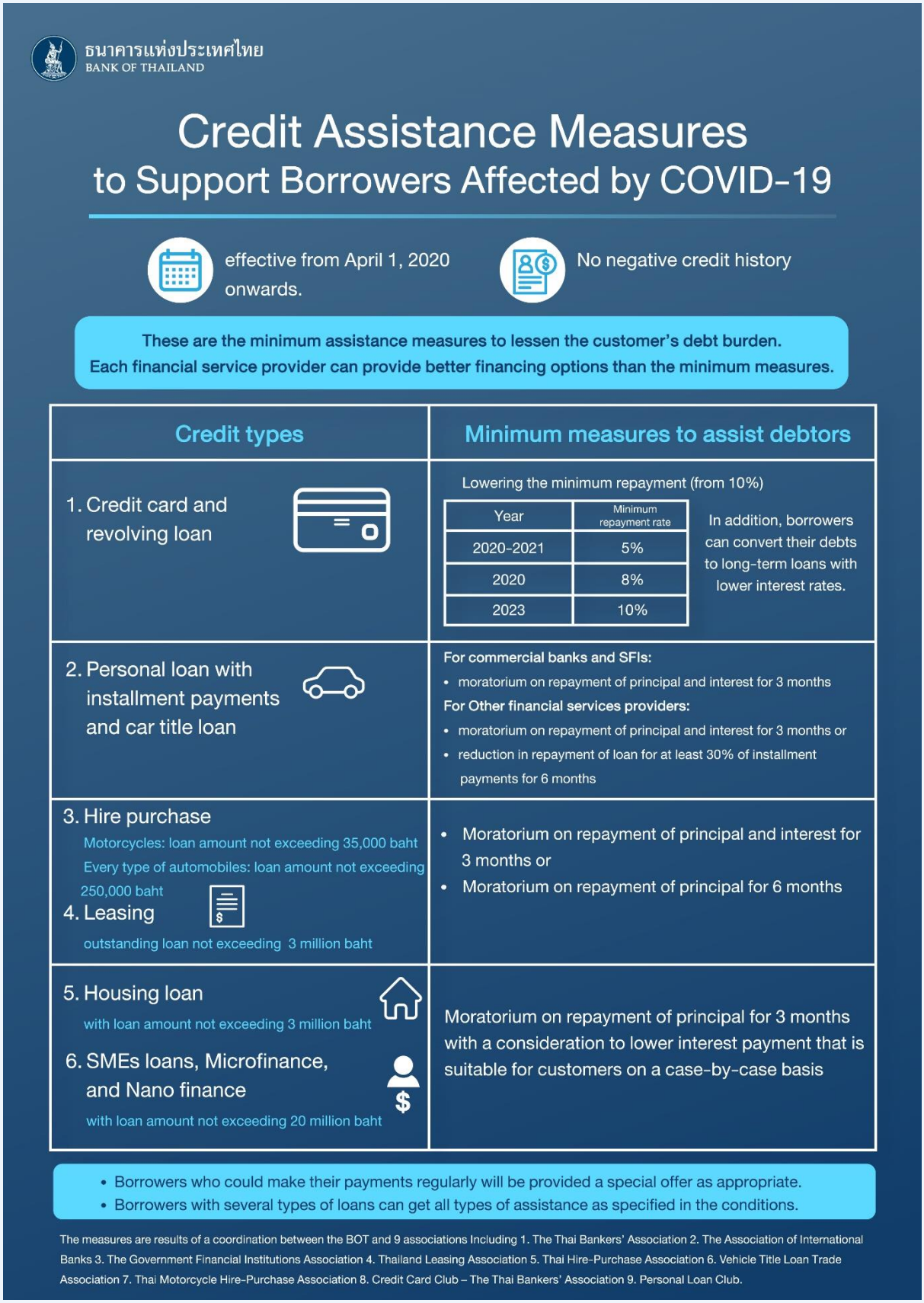
The Committee would stand ready to use additional policy tools in an appropriate and timely manner. Such tools included the policy rate and other financial measures.

Looking ahead, the Committee would monitor developments of economic growth, inflation, and financial stability, together with associated risks, especially the impact of the COVID-19 outbreak, in deliberating monetary policy going forward. The Committee would stand ready to use additional policy tools in an appropriate and timely manner. Such tools included the policy rate and other monetary measures that would help enhance the policy rate transmission as well as loan extension by financial institutions to reach the target groups.

[16/] BOT Press Release NO. 11/2020 "Increase of repatriation threshold to 1 million U.S. dollars"

Credit assistance measures to support borrowers

Chart 1 Loan repayment measures to assist borrowers affected by COVID-19



The Bank of Thailand (BOT) coordinated with financial institutions and other financial service providers to assist borrowers affected by the COVID-19.

In recent periods, the Thai economy has been continuously affected by negative risk factors, including trade tensions, the drought, as well as increasingly severe COVID-19 outbreak. On February 28, 2020, the BOT, in collaboration with financial institutions, issued proactive measures to assist borrowers of all types of business loans and households affected by the outbreak. These measures will be effective for a period of 2 years between January 1, 2020 to December 31, 2021 and aim at supporting liquidity provision and accelerating debt restructuring process. Examples include provision of additional working capital, a loan repayment holiday, an extension of loan repayment period, interest and fee reduction, minimum repayment reduction for credit card loans to below 10 percent, and a temporary increase in the credit limit of personal loans.

The BOT has been coordinating closely with financial institutions and other financial service providers and agreed that the impact of the COVID-19 would become more severe and widespread. Thus, all parties deemed it urgent to adopt a set of minimum measures to support retail and SME borrowers with the potential to repay their loans and whose loans are not past due for longer than 90 days (NPL). Such measures will become effective from the debt repayment period of April 1, 2020 onwards. Details are summarized as in Chart 1, where each financial service provider could offer relief packages that are better than the minimum measures.

The BOT relaxed regulations to further encourage financial institutions to provide liquidity to affected borrowers.

The BOT temporarily relaxed liquidity-related regulations[17/] in order to reduce credit costs of financial institutions that increased from liquidity provision and debt restructuring for affected borrowers. This would enable financial institutions to assist borrowers more extensively and faster. Details of relaxed regulations are as follows.

(1) Pre-emptive debt restructuring for borrowers with signs of debt repayment problems but are not yet classified as NPL. Debt restructuring for these borrowers will not be classified as trouble debt restructuring (TDR), thus not affecting borrowers' credit history. This will remain a permanent measure.

(2) Classification of borrowers entitled to the relief measures:

(2.1) Borrowers under pre-emptive debt restructuring can be immediately classified as normal borrowers.

(2.2) NPL borrowers, who can make repayments in accordance with the debt restructuring agreement for 3 consecutive months, can be classified as normal borrowers (from previously having to make repayment for 12 consecutive months).

(2.3) Borrowers obtaining additional working capital can be immediately classified as normal borrowers.

(3) Provision not required for credit lines of borrowers that are not yet activated.

[17/] For the duration of 2 years starting from January 1, 2020 until December 31, 2021

(4) Period for holding non-performing assets (NPA) obtained from debt restructuring for financial institutions was extended to 10 years (from previously 5 years).

(5) Relationship managers (RM) can provide assistance regarding debt restructuring to their clients with prudent internal controls.

(6) Financial institutions must report credit conditions and progress in assisting these borrowers to the BOT on a monthly basis so that the BOT can continuously monitor the situation.

Related government agencies launched measures to support debt restructuring and additional credit provision

(1) Tax relief measures: exemption from personal and corporate income taxes for borrowers' income earned from forgiven debt; exemption from income tax, value added tax (VAT), and specific business tax (SBT), stamp duties for both borrowers and creditors imposed on income earned from transfer of assets, sales of goods or provision of services, and any instrument executed in relation to debt restructuring, as well as reduction of fees for registration and transfer of property from 2 to 0.01 percent.

(2) Financial statements of financial institutions on classification and provision: Auditors shall provide opinion without bias in compliance with relevant accounting and supervisory standards of the BOT.

(3) The Thai Credit Guarantee Corporation (TCG) launched additional credit guarantee facilities, including "The Funding Thai SMEs for Thailand" program with a guaranteed amount of 60 billion baht, and the 8th phase of Portfolio Guarantee Scheme (PGS8) with a guaranteed amount of 50 billion baht. These will somewhat help reduce credit costs of financial institutions. In addition, TCG will extend the maturity of credit guarantee period for SMEs already under PGS5 – PGS7 for another five years in order to facilitate debt restructuring. There will also be an adjustment in claiming procedures that can be done without borrowers having to enter into a litigation, and an expansion in guarantee coverage.

These measures will only be successful if financial institutions and other financial service providers play an active role, while more borrowers cooperate in accelerating debt restructuring.

The BOT, financial institutions, and other financial service providers aim to ease interest burdens of those affected by the outbreak in order to enhance their liquidity by urgently implementing minimum measures and additional debt restructuring measures to have a noticeable impact. The BOT has relaxed regulations to reduce credit costs of financial institutions arising from proactive measures to support borrowers. Such credit assistance measures for borrowers will reinforce financial and fiscal measures gradually implemented previously in order to reduce funding costs for financial institutions. Such measures include, for instance, a more accommodative monetary policy by cutting the policy rate to a historically low level, and soft loan worth 1.5 trillion baht funded by the Government Saving Bank. However, these support measures for borrowers will only be successful if financial institutions and other financial service providers play an active role, while more borrowers cooperate in accelerating debt restructuring. The BOT will continue to monitor and push forward measures to assist borrowers so that borrowers affected by the outbreak would be able to pass through this difficult time.

Measures to stabilize the Thai financial markets

In March 2020, the situation on the COVID-19 spread intensified and raised concerns among global financial markets. Asset prices were highly volatile. Investors in many countries thus continuously sold off their assets, resulting in tightening global liquidity. This led to irregular functioning of global financial markets and subsequently affected the Thai financial markets, especially liquidity in the bond market. This was driven by sell-off by foreign investors and some mutual funds following a fast redemption of investment units of some investors. This resulted in liquidity shortages of those mutual funds, forcing them to liquidate their bonds before maturities at lower than fair value prices. Most of these bonds were high-quality bond including government and investment-grade corporate bonds. In this situation, investment unit holders in other mutual funds thus received lower returns than they should. This could result in accelerated redemption of investment units in other mutual funds, consequently disrupting the functioning of mutual funds and the Thai bond markets.

The Bank of Thailand (BOT) addressed the liquidity situation to support the well-functioning of the bond market. The BOT purchased government bonds amounting more than 100 billion baht during March 13-20, 2020, and reduced and cancelled parts of the BOT bond issuances in order to increase liquidity in the government bond market. In addition, **the BOT, the Ministry of Finance, and the Securities and Exchange Commission (SEC) have been closely monitoring the situation and deemed it necessary to implement additional measures to stabilize the Thai financial markets on March 22, 2020.** These measures would stop the liquidity problem in the financial markets from spreading further. The three measures are summarized as follows.[18/]

1. Mutual Fund Liquidity Facility (MFLF) The BOT will provide liquidity through the repurchase agreement or repo transactions charging a special rate based on the policy rate minus 0.5 percent for financial institutions providing liquidity support for money market funds (MMFs) and daily fixed income funds (Daily FI). On March 24, 2020, the BOT extended the scope of this liquidity facility to cover all MMFs and Daily FI funds. Later on March 30, 2020, the BOT further extended the scope of the facility in order to increase flexibility and efficiency by allowing more types of assets to be used as collateral in repo transactions with the BOT for the liquidity support.

2. Corporate Bond Liquidity Stabilization Fund (BSF) was under consideration for this Fund to be urgently established. The Fund's objective is to provide bridge financing to purchase bonds issued by high-quality firms maturing during 2020-2021. Nevertheless, firms requesting for liquidity support from the BSF will be charged a higher interest rate than market rates. These firms should have raised the majority of their funding needs through other means, such as commercial bank loans or capital increase, have a clear long-term financing plan, and meet other conditions as set out by the BSF's investment committee. In addition, if the issuers simultaneously offer secured bonds to the general public, the bonds that the BSF will invest in must also be secured with collaterals no inferior than those pledged on the bonds sold to the general public.

[18/] Joint Statement on the Measures to Stabilize the Financial Markets (March 22, 2020)

3. The BOT stands ready to continue its liquidity provision in the bond market through government bond purchases and consider additional measures in order to enable an efficient functioning of the Thai financial markets. This would ensure sufficient liquidity to provide a funding channel for the economic system and preserve financial stability of the country as a whole.

Measures to stabilize the Thai financial markets abovementioned will help enhance liquidity in the financial markets, ensure the normal functioning of the bond markets amid turbulences in the global financial markets, and restore confidence of both Thai and foreign investors in the bond market. All related agencies, both in the public and private sectors, will be closely monitoring developments in the financial markets and stand ready to coordinate in implementing additional measures. This would ensure that the Thai financial markets will continue to function normally and efficiently.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2018	2019	2018	2019			
			Q4	Q1	Q2	Q3	Q4
GDP growth	4.2	2.4	3.8	2.9	2.4	2.6	1.6
Production							
Agriculture	5.5	0.1	1.3	1.7	-1.4	2.7	-1.6
Non-agriculture	4.0	2.6	4.1	3.0	2.8	2.5	2.0
Manufacturing	3.2	-0.7	3.8	0.2	0.1	-0.8	-2.3
Construction	2.4	2.0	3.0	3.0	3.4	2.7	-1.9
Wholesales and retail trade	6.6	5.7	7.3	6.7	5.5	5.3	5.2
Accommodation and Food Service	7.6	5.5	5.1	5.0	3.6	6.7	6.8
Transport and storage	4.4	3.4	4.0	3.7	3.0	3.1	3.9
Information and Communication	7.6	8.8	7.3	7.1	8.8	8.2	10.8
Financial intermediation	3.4	2.7	1.9	2.0	1.8	3.8	3.4
Real estate and renting	5.7	3.4	5.4	5.0	3.3	2.2	3.0
Expenditure							
Domestic demand	4.0	3.3	4.3	4.1	3.4	3.4	2.3
Private consumption	4.6	4.5	5.1	4.8	4.7	4.3	4.1
Private investment	4.1	2.8	5.7	4.3	2.1	2.3	2.6
Government consumption	2.6	1.4	2.2	3.5	1.5	1.7	-0.9
Public investment	2.9	0.2	-0.3	0.0	1.5	3.7	-5.1
Imports of goods and services	8.3	-4.4	4.6	0.1	-3.4	-5.9	-8.3
imports of goods	7.9	-5.5	3.8	-2.8	-3.7	-6.8	-8.6
imports of services	9.9	0.0	7.7	13.0	-2.1	-2.1	-7.1
Exports of goods and services	3.3	-2.6	1.3	-3.5	-4.0	0.6	-3.6
exports of goods	3.8	-3.6	1.1	-4.4	-4.6	-0.1	-5.1
exports of services	2.0	0.4	2.0	-1.1	-1.7	3.2	1.1
Trade balance (billion, U.S. dollars)	22.4	26.6	4.6	6.7	6.0	8.0	5.9
Current account (billion, U.S. dollars)*	28.5	37.9	6.3	12.4	5.5	9.7	10.3
Financial account (billion, U.S. dollars)*	-14.9	-12.9	-2.5	-5.1	-2.5	-1.1	-4.1
International reserves (billion, U.S. dollars)	205.6	224.3	205.6	212.2	215.8	220.5	224.3
Unemployment rate (%)	1.1	1.1	0.9	0.9	1.0	1.0	1.0
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.0	0.9	0.9	1.0	1.0

Note: *Data may be subjected to revisions and compilation method
Source: NESDC, National Statistical Office, and Bank of Thailand

Financial Stability Dashboard

Indicators	2018	2019	2018	2019				2020	
			Q4	Q1	Q2	Q3	Q4	Jan	Feb
1. Financial market sector									
Bond market									
Bond spread (10 years - 2 years)	0.7	0.3	0.7	0.7	0.4	0.1	0.3	0.2	0.3
Equity market									
SET index (end of period)	1,563.9	1,579.8	1,563.9	1,638.7	1,730.3	1,637.2	1,579.8	1,514.1	1,340.5
Actual volatility of SET index[1/]	12.1	10.2	14.0	8.4	7.8	10.7	10.2	16.5	27.2
Price to Earnings ratio (P/E ratio) (tim	14.8	19.4	14.8	17.1	18.6	18.7	19.4	18.4	15.7
Exchange rate market									
Actual volatility of Thai baht (%annuali	4.6	4.1	4.8	4.6	4.5	4.6	2.7	3.8	5.3
Nominal Effective Exchange Rate (NE	115.6	123.2	117.0	119.9	120.9	125.0	126.8	125.1	123.0
Real Effective Exchange Rate (REER	107.2	112.8	108.3	110.1	111.3	114.6	115.3	113.0	111.2
2. Financial institution sector[3/]									
Minimum Lending Rate (MLR)[4/]	6.28	6.08	6.28	6.28	6.28	6.28	6.08	6.08	5.98
12-month fixed deposit rate[4/]	1.37	1.33	1.37	1.42	1.46	1.45	1.33	1.29	0.98
Capital adequacy									
Capital funds / Risk-weighted asset (%	18.3	19.3	18.3	18.2	18.0	18.9	19.3	19.2	n.a.
Earning and profitability									
Net profit (billion, Thai baht)	207.2	270.8	49.5	57.1	60.8	96.533	56.4	n.a.	n.a.
Return on assets (ROA) (times)	1.1	1.4	1.1	1.2	1.3	2.0	1.4	n.a.	n.a.
Liquidity									
Loan to Deposit and B/E (%)	98.2	95.5	98.2	96.8	97.9	97.1	95.5	96.5	96.3
3. Household sector									
Household debt to GDP (%)	78.4	79.5	78.4	78.1	78.1	78.5	79.5	n.a.	n.a.
Financial assets to debt (times)	2.4	n.a.	2.4	2.5	n.a.	n.a.	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Consumer loans	2.7	2.9	2.7	2.8	2.7	2.8	2.9	n.a.	n.a.
Housing loans	3.2	3.7	3.3	3.4	3.3	3.5	3.7	n.a.	n.a.
Auto leasing	1.7	1.9	1.7	1.7	1.8	1.9	1.9	n.a.	n.a.
Credit cards	2.3	2.4	2.3	2.7	2.5	2.7	2.4	n.a.	n.a.
Other personal loans	2.5	2.3	2.5	2.6	2.4	2.4	2.3	n.a.	n.a.
4. Non-financial corporate sector[5/]									
Operating profit margin (OPM) (%)	7.5	6.9	6.5	7.5	5.9	7.4	6.9	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.8	0.8	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.4	4.1	5.4	5.2	3.8	4.8	4.1	n.a.	n.a.
Current ratio (times)	1.6	1.6	1.6	1.6	1.6	1.6	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Large businesses	1.6	1.5	1.6	1.5	1.6	1.6	1.5	n.a.	n.a.
SMEs	4.6	4.6	4.6	4.7	4.5	4.8	4.6	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2018	2019	2018	2019				2020	
			Q4	Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector									
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)									
Total	73,361	70,876	21,623	19,710	15,983	17,141	18,042	3,261	n.a.
Single-detached and semi-detache	15,912	16,167	4,610	4,446	3,460	4,136	4,125	797	n.a.
Townhouses and commercial build	25,042	24,763	7,495	7,290	5,719	6,129	5,625	1,206	n.a.
Condominiums	32,407	1,718	9,518	7,974	6,804	6,876	8,292	1,258	n.a.
Number of new housing units launched for sale (Bangkok and Vicinity) (units)									
Total	124,977	116,933	35,861	29,498	25,999	27,182	34,254	5,368	6,140
Single-detached and semi-detache	18,311	19,423	5,050	4,458	4,213	5,014	5,738	514	1,134
Townhouses and commercial buildi	32,349	31,423	10,385	5,236	7,940	9,322	8,925	1,546	3,025
Condominiums	74,317	66,087	20,426	19,804	13,846	12,846	19,591	3,308	1,981
Housing price index (2009 = 100)									
Single-detached houses (including lan	138.6	144.0	139.0	141.9	142.9	145.0	146.3	148.4	n.a.
Townhouses (including land)	149.5	156.4	151.1	156.6	154.7	154.6	159.8	161.8	n.a.
Condominiums	180.9	182.3	185.9	181.6	177.0	179.0	191.6	197.3	n.a.
Land	173.6	172.8	177.6	175.5	168.1	167.1	180.6	180.9	n.a.
6. Fiscal sector									
Public debt to GDP (%)	41.8	41.2	41.8	41.8	41.3	41.1	41.2	41.3	n.a.
7. External sector									
Current account balance to GDP (%)	5.6	7.0	4.8	9.3	4.1	7.1	7.2	n.a.	n.a.
External debt to GDP (%)[6/]	35.4	34.3	35.4	35.0	34.7	34.1	34.3	n.a.	n.a.
External debt (billion, U.S. dollars)	162.4	172.1	162.4	164.5	166.7	167.2	172.1	173.1	170.3
Short-term (%)	39.0	34.8	39.0	38.5	37.5	35.4	34.8	35.0	34.5
Long-term (%)	61.0	65.2	61.0	61.5	62.5	64.6	65.2	65.0	65.5
International reserves / Short-term ex	3.2	3.7	3.2	3.4	3.5	3.7	3.7	3.8	3.9

Note:

[6/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2019	2020	2021						
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 15	0	0	0	0	0	0	1	0	0
12-15	0	0	0	0	0	0	4	2	1
9-12	0	0	0	0	0	2	11	6	4
6-9	0	0	0	0	0	7	18	14	10
3-6	0	0	0	0	2	15	20	20	17
0-3	100	4	0	2	8	19	17	20	21
(-3)-0	0	31	0	9	16	18	12	16	19
(-6)-(-3)	0	38	7	19	18	15	8	11	14
(-9)-(-6)	0	20	28	21	17	11	4	6	8
(-12)-(-9)	0	6	32	18	14	7	2	3	4
(-15)-(-12)	0	1	21	14	10	4	1	1	2
(-18)-(-15)	0	0	9	9	7	2	0	0	1
(-21)-(-18)	0	0	3	5	4	1	0	0	0
< (-21)	0	0	1	4	4	0	0	0	0

Table: Probability distribution of headline inflation forecast

Percent	2019	2020	2021						
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 3.0	0	0	0	0	1	1	9	11	12
2.5-3.0	0	1	0	0	1	1	5	6	6
2.0-2.5	0	3	0	1	2	2	7	7	7
1.5-2.0	0	7	1	2	3	3	8	9	8
1.0-1.5	0	14	2	3	5	5	9	9	9
0.5-1.0	0	20	4	5	6	7	10	10	9
0.0-0.5	100	21	6	7	8	8	10	9	9
(-0.5)-0.0	0	17	9	9	10	10	9	9	8
(-1.0)-(-0.5)	0	10	11	11	10	10	8	8	7
(-1.5)-(-1.0)	0	5	13	12	11	10	7	6	6
(-2.0)-(-1.5)	0	2	13	12	10	10	5	5	5
(-2.5)-(-2.0)	0	0	12	11	9	9	4	4	4
(-3.0)-(-2.5)	0	0	10	9	7	7	3	3	3
< -3.0	0	0	17	18	16	16	5	4	6

Table: Probability distribution of core inflation forecast

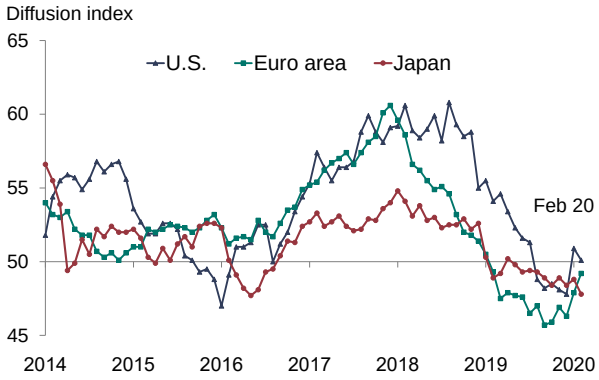
Percent	2019	2020	2021						
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 2.5	0	0	0	0	0	0	0	0	0
2.0-2.5	0	0	0	0	0	0	0	1	1
1.5-2.0	0	0	0	0	1	1	1	4	4
1.0-1.5	0	8	4	2	3	2	5	9	9
0.5-1.0	0	38	16	8	7	7	10	14	14
0.0-0.5	100	40	31	17	13	13	16	16	15
(-0.5)-0.0	0	12	28	24	18	18	17	15	14
(-1.0)-(-0.5)	0	1	15	22	19	18	16	13	12
(-1.5)-(-1.0)	0	0	5	15	16	16	13	10	10
(-2.0)-(-1.5)	0	0	1	8	11	12	9	7	7
(-2.5)-(-2.0)	0	0	0	3	6	7	6	5	5
(-3.0)-(-2.5)	0	0	0	1	3	4	3	3	3
(-3.5)-(-3.0)	0	0	0	0	1	2	2	2	2
< -3.5	0	0	0	0	0	1	1	1	2

4.2 Chart Pack

Global Economy

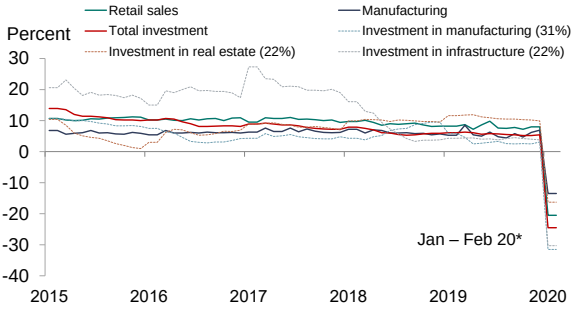
The COVID-19 pandemic caused major advanced economies to enter into recession in 2020, while Chinese and Asian economies would experience a slower economic growth. The Chinese economy was severely affected, as reflected by a sharp contraction of the economic indicators. Governments in various countries worldwide promptly released measures to alleviate the impact of the COVID-19.

Manufacturing Purchasing Manager Index



Sources: Bloomberg and Eurostat

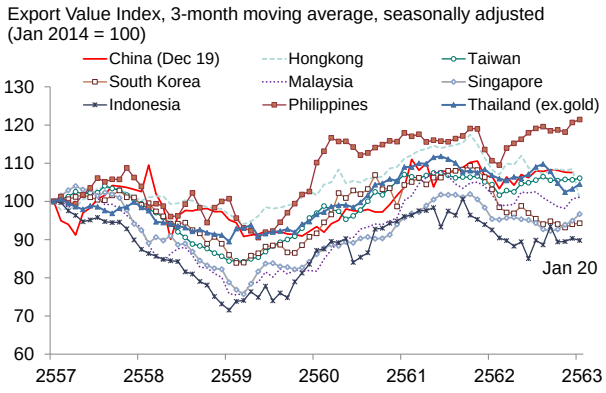
China's economic indicators (Change from the same period last year)



Note: ( ) denotes share to total investment
* last data is monthly average between January and February 2020

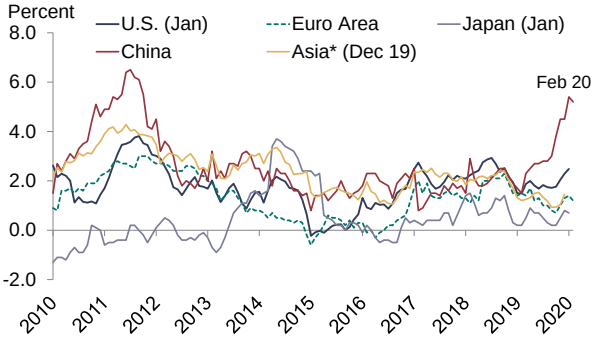
Source: CEIC

Asian exports



Source: CEIC

Headline inflation of Thailand's major trading partners



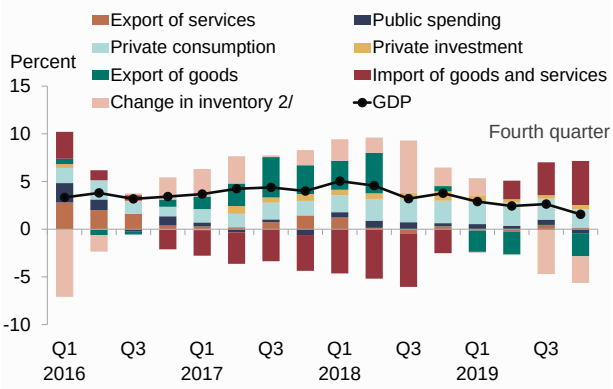
Note: *Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore, and Taiwan

Source: CEIC

Thai economy

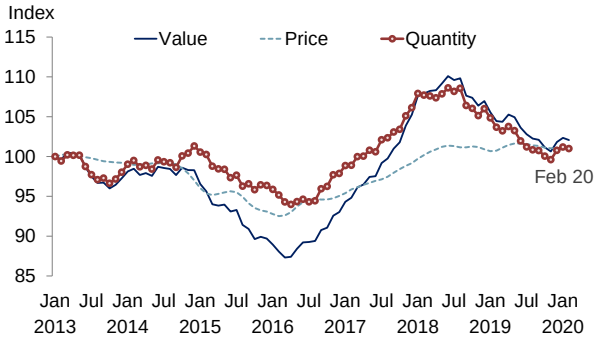
The Thai economy in the fourth quarter of 2019 expanded at a slower pace due to both domestic and external demand. The delayed enactment of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020) had a larger-than-expected impact on public expenditure. The Thai economy in the first quarter of 2020 was projected to contract, mainly as tourism and merchandise exports were severely affected by escalating COVID-19 outbreak situation.

Contribution to Thailand's GDP growth[1/]



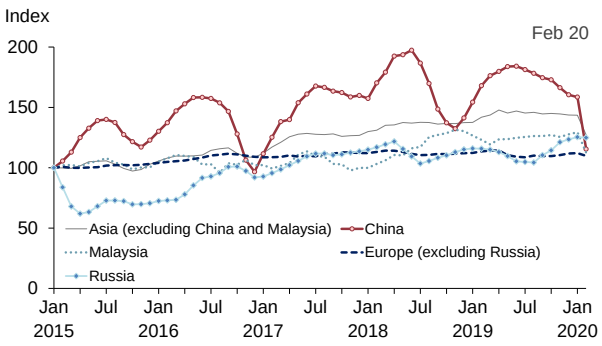
Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Council, calculations by the Bank of Thailand

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



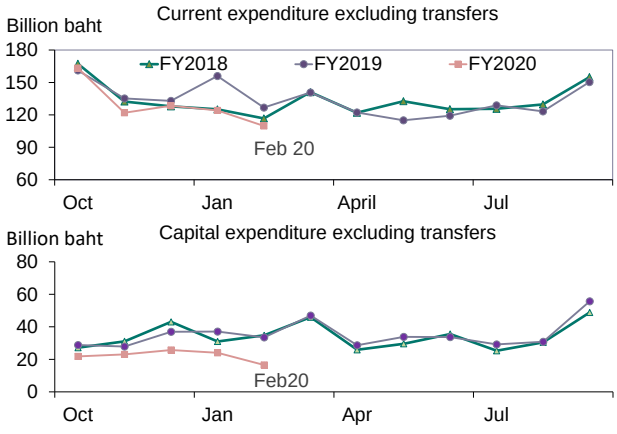
Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Index of foreign tourists classified by nationality
(three-month moving average, seasonally adjusted; January 2015 = 100)



Source: Department of Tourism

Public spending by central government

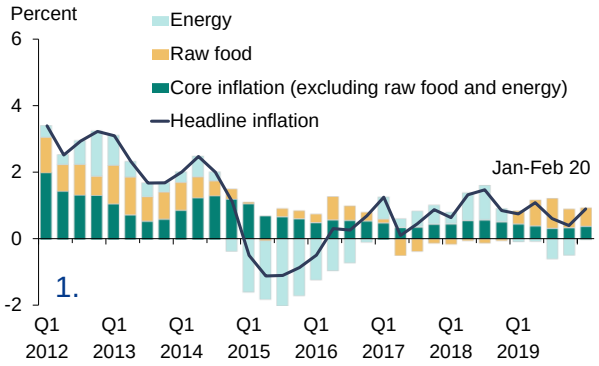


Source: Comptroller General's Department, Fiscal Policy Office

Inflation

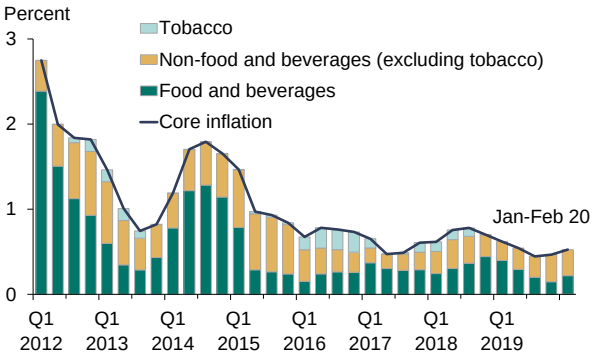
Headline inflation in January-February increased due to the base effect of energy prices, but was projected to decline in March in line with global crude oil prices. Core inflation increased in tandem with prices of seasonings and condiments. Short-term inflation expectations declined slightly while long-term inflation expectations remain largely unchanged.

Contribution to headline inflation



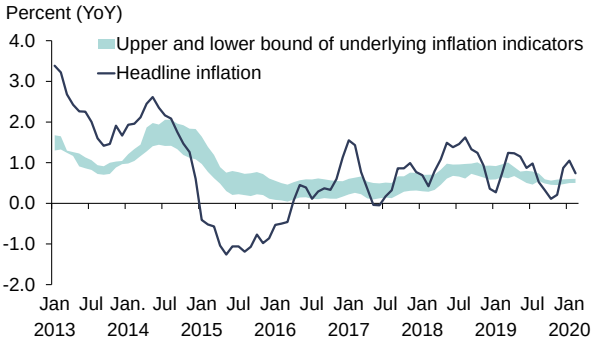
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

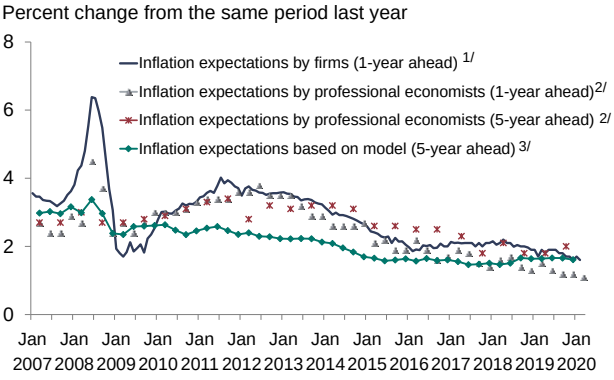
Underlying inflation indicators



Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

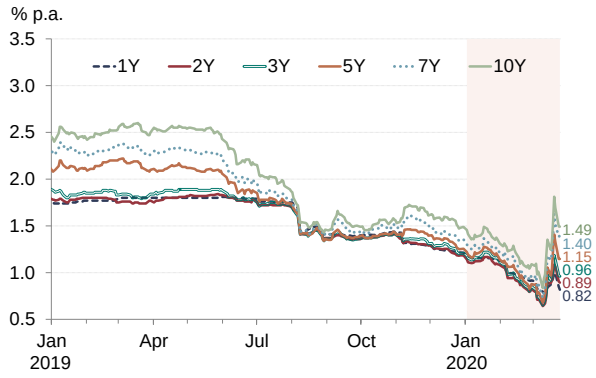


Source: 1/ Business Sentiment Survey (BSI) by the Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model using bond yield and macroeconomic data

Financial conditions

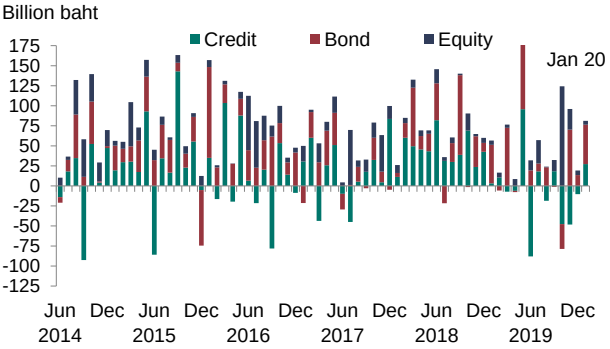
Short-term government bond yields and commercial banks' interest rates fell in line with the policy rate. Long-term government bond yields declined in tandem with U.S. treasury yields, before rising rapidly towards the end of the quarter on the back of concerns over the COVID-19 situation. Business credits expanded as a result of corporate bond issuance. The baht against the U.S. dollar depreciated due to the COVID-19 and the Committee's policy rate cut at the special meeting.

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of 24 Mar 2020)

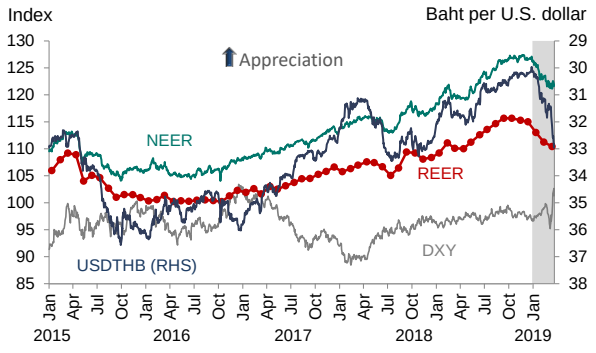
Total corporate financing by instrument*



Note: *Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding those issued by commercial banks, and newly issued equities.

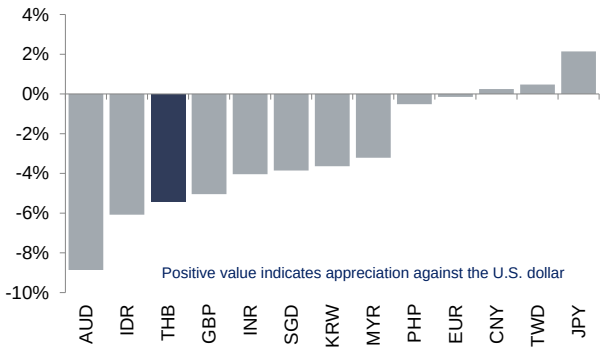
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 24 March 2020)

Currency movements vis-a-vis the U.S. dollar (Avg. in Mar 2020 compared to avg. in Dec 2019)

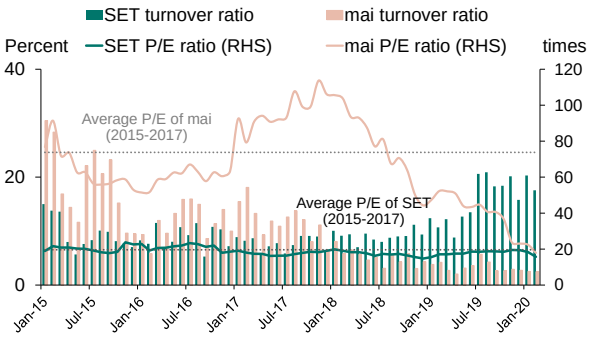


Source: Bank of Thailand and Reuters (data as of 24 March 2020)

Stability: financial markets

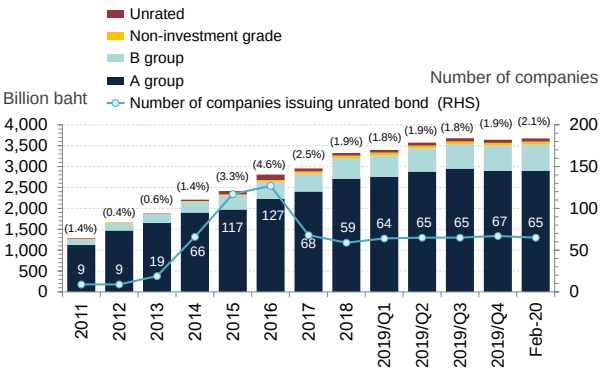
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) and the Market for Alternative Investment (mai) decreased in tandem with stock market indices worldwide. The share of unrated bond issuance remained stable.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of February 2020)

Corporate bonds outstanding

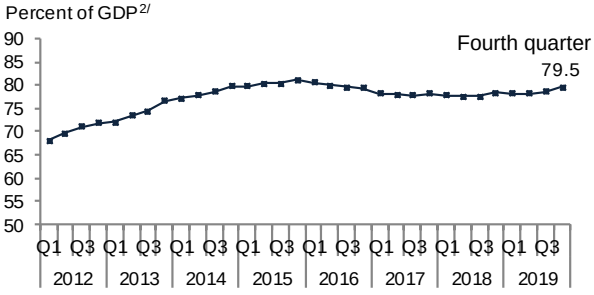


Note: ( ) represents percentage of unrated bonds to total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

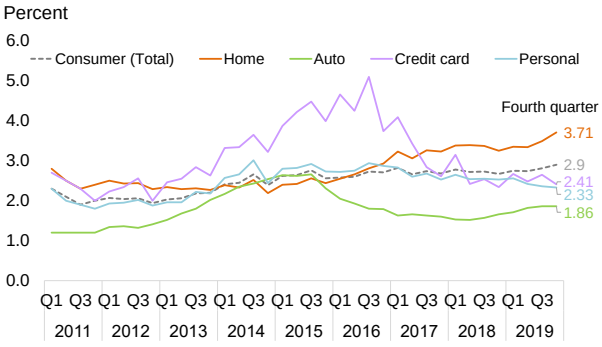
The ratio of household debt to GDP in the fourth quarter of 2019 remained high and increased slightly from the previous quarter. Credit quality of households was expected to deteriorate as in response to income shocks, where the NPL ratio of households increased from the previous quarter mainly in line with deteriorating quality of mortgage loans and personal loans with housing as collateral.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

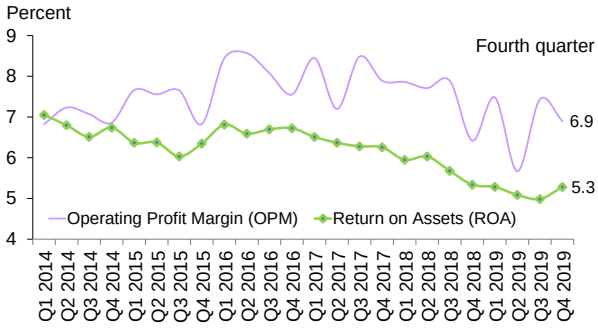


Source: Bank of Thailand

Stability: corporate sector

Overall vulnerabilities in the corporate sector increased, especially for SMEs, as reflected by the sustained decline of operating profit margin (OPM) and negative interest coverage ratio (ICR) of some businesses. Credit quality of business loans deteriorated across almost all categories, having extended to include larger enterprises. The NPL ratio of SME loans in the fourth quarter of 2019 remained high at 4.6 percent.

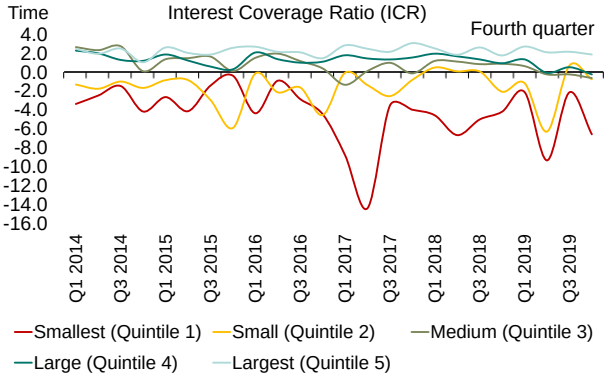
Operating Profit Margin (OPM) and Return on Assets (ROA)*



Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.

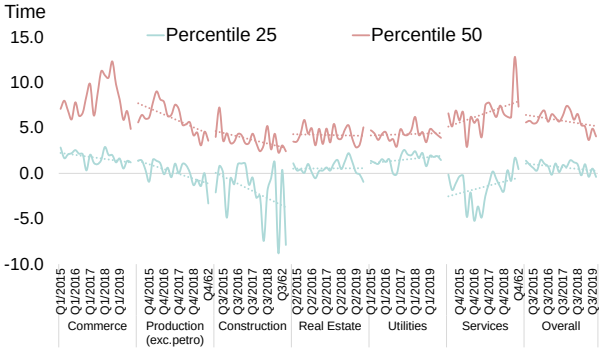
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



Source: Stock Exchange of Thailand, calculation by Bank of Thailand

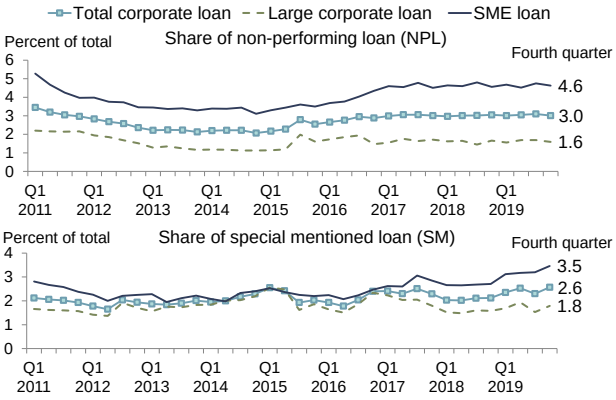
Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

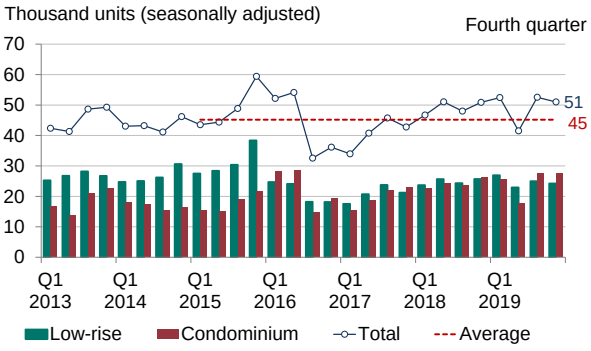


Source: Bank of Thailand

Stability: real estate

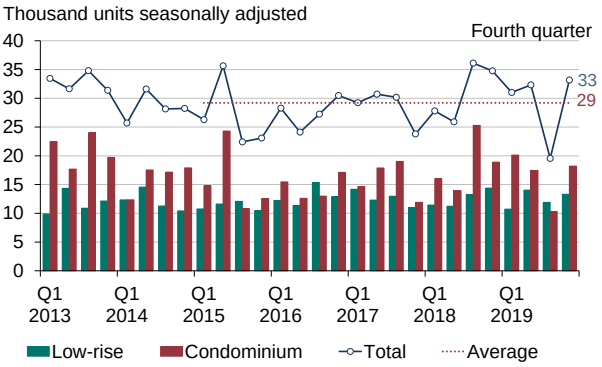
Oversupply in the real estate market increased, corresponded to the longer time-to-go caused by (1) weaker demand for residences from domestic and foreign buyers as evident from declining ownership transfers, and (2) new residential project launches despite weakening demand. The impact of the relaxed loan-to-value (LTV) measure announced in early 2020 would need to be monitored going forward.

Residential Transfer units in Bangkok and its vicinity



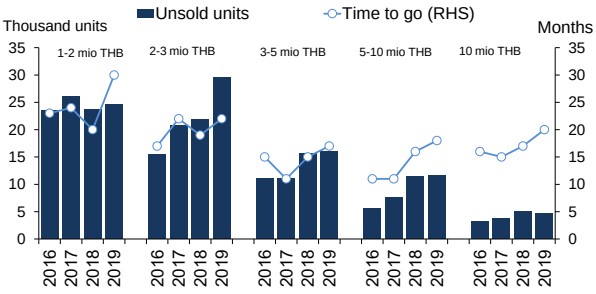
Source: Real Estate Information Center
Note: *Average during 2014-2018

New residential projects launched in Bangkok and its vicinity



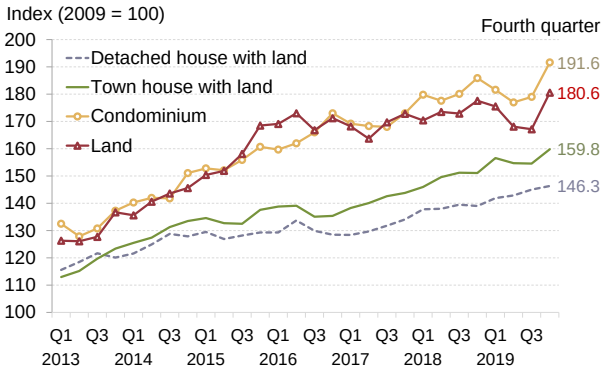
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand
Note: *Average during 2014-2018

Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.
Source: AREA and calculation by the Bank of Thailand

Real estate price indices

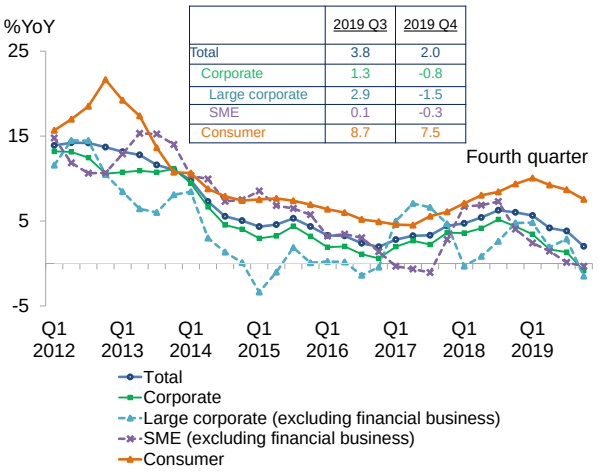


Note: Calculation based on commercial bank loan data
Source: Bank of Thailand

Stability: financial institutions

Financial institutions maintained robust financial positions, as reflected by adequately high levels of commercial banks' capital buffers to cushion against risks of credit quality deterioration. In the fourth quarter of 2019, commercial bank credits expanded at a slower pace in line with economic conditions. Overall NPL ratio remained largely unchanged from the previous quarter.

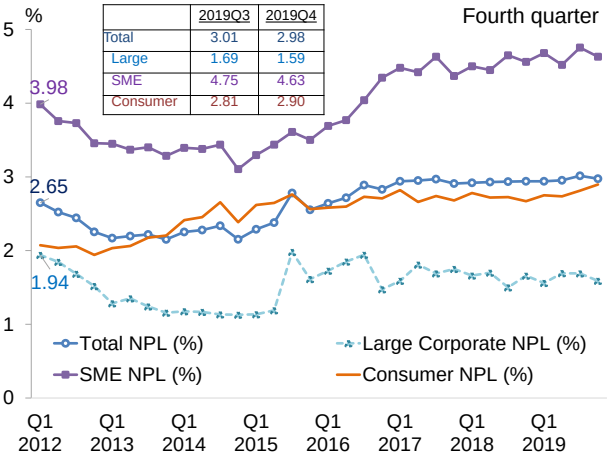
Credit growth in the commercial bank system



	2019 Q3	2019 Q4
Total	3.8	2.0
Corporate	1.3	-0.8
Large corporate	2.9	-1.5
SME	0.1	-0.3
Consumer	8.7	7.5

Source: Bank of Thailand

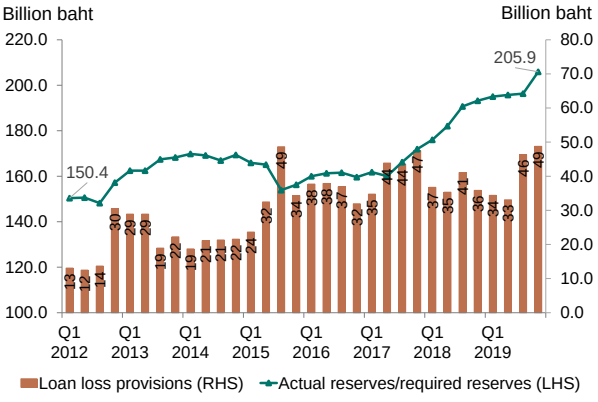
Non-performing loan (NPL)



	2019Q3	2019Q4
Total	3.01	2.98
Large	1.69	1.59
SME	4.75	4.63
Consumer	2.81	2.90

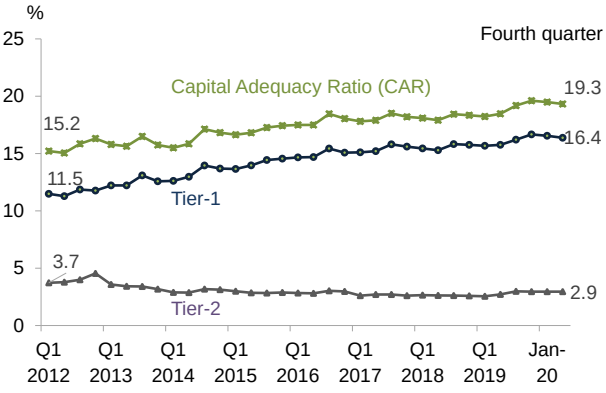
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

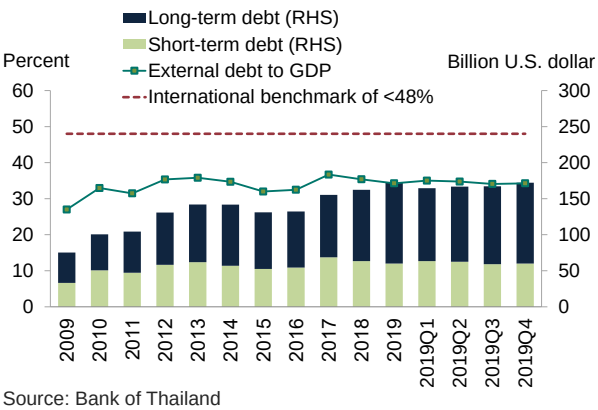


Source: Bank of Thailand

Stability: external positions

Thailand's external stability remained sound, as reflected in high levels of international reserves. The ratio of external debt to GDP was below an international benchmark. This would help the Thai economy stay resilient against volatilities in the global financial market.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

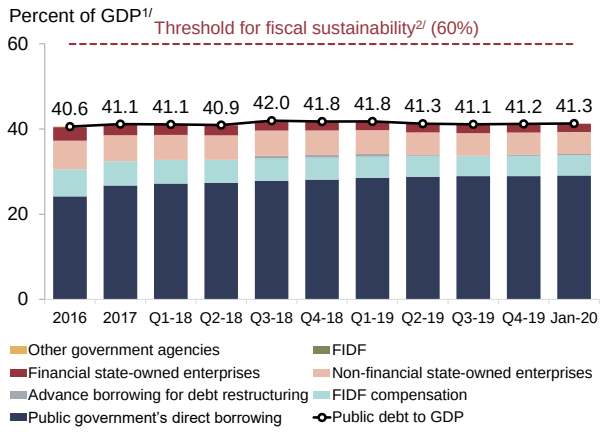


Source: Bank of Thailand

Stability: fiscal sector

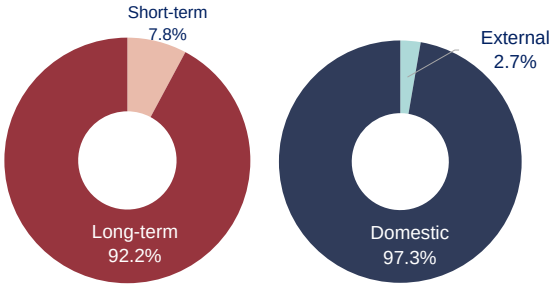
Fiscal stability remained sound. The ratio of public debt to GDP remained below the fiscal sustainability threshold.

Public debt to GDP



Note: 1/ Calculated from GDP by Chain Volume Measure method
2/ Set by Monetary and Fiscal Policy Committee by the authority of State Fiscal and Financial Disciplines Act, B.E. 2561 (2018)
Source: Public Debt Management Office

Outstanding debt as of January 2020



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office



Pursuing Sustainable Economic Well-Being

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